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Diffusion-Based Generative Models for Reconstructing Natural Speech from EEG Brain Signals

MASTER'S THESIS

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STUDENT DECLARATION

I, *Fatima Ezzahra Akaaboune*, the undersigned, hereby declare that the present Master's thesis work has been prepared by myself and without any unauthorized help or assistance. Only the specified sources (references, tools, etc.) were used. All parts taken from other sources word by word, or after rephrasing but with identical meaning, were unambiguously identified with explicit reference to the sources utilized. In the Appendix, I have clearly indicated whether I used artificial intelligence tools in preparing the thesis; if so, I have specified the manner and extent of their use in the table. I acknowledge that I bear full responsibility for any content generated by artificial intelligence – regardless of the extent of its use.

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Budapest, 7 June 2026



Fatima Ezzahra Akaaboune
student

Résumé

La communication cerveau-ordinateur offre un potentiel transformateur pour aider les personnes atteintes de troubles graves de la parole. Ce mémoire étudie la reconstruction directe de la parole à partir d'enregistrements intracrâniens, dans le but de développer des modèles capables de traduire l'activité neuronale en un signal audio intelligible. Afin de garantir la robustesse informatique et la généralisation inter-modalités, nous utilisons deux jeux de données fondamentalement distincts : le jeu de données UCSF DANDI (ECoG de surface à haute densité capturant des syllabes en anglais) et le jeu de données de Maastricht *SingleWordProductionDutch-iBIDS* (sEEG profond et clairsemé capturant des mots en néerlandais).

Pour modéliser la relation hautement non linéaire entre l'activité neuronale et les caractéristiques acoustiques, nous avons conçu un pipeline d'apprentissage automatique en deux étapes. Tout d'abord, les enveloppes de fréquences high-gamma ont été extraites et synchronisées avec des spectrogrammes Mel cibles générés par synthèse vocale (TTS). Ensuite, un modèle de régression linéaire de base a été utilisé comme mécanisme de filtrage pour vérifier l'intégrité du signal, suivi d'un modèle de Diffusion. Par un processus itératif de débruitage des spectrogrammes conditionné par les caractéristiques neuronales, l'architecture générative de diffusion réussit à synthétiser des structures acoustiques complexes que les modèles linéaires ne parviennent pas à résoudre.

Les résultats expérimentaux démontrent que le modèle de diffusion atteint un alignement temporel fort sur les deux modalités, avec des corrélations de Pearson atteignant des pics de 0,97 sur le jeu de données sEEG et de 0,91 sur le jeu de données ECoG. Cependant, l'évaluation qualitative a révélé une divergence critique entre les métriques statistiques et la qualité audio perceptuelle. Alors que la modalité sEEG a capturé une variance spectrale suffisante pour produire un signal audio intelligible, les résultats démontrent que la cartographie de la cinématique ECoG de surface — bien qu'atteignant des frontières temporelles très précises — produit des prédictions qui sont trop rigides pour synthétiser les gradients de fréquences internes. Par conséquent, les formes d'onde ECoG ont subi une interférence destructive et un effondrement sévère de l'amplitude lors de la reconstruction de la phase par l'algorithme de Griffin-Lim. Ce travail fournit un cadre systématique et mathématiquement robuste pour le décodage neuronal, soulignant comme conclusion scientifique majeure la nécessité d'intégrer des vocodeurs neuronaux pour les futures neuroprothèses.

Abstract

Brain-computer communication offers transformative potential for assisting individuals with severe speech impairments. This thesis investigates the direct reconstruction of speech from intracranial recordings, aiming to develop models that can accurately translate neural activity into intelligible audio. To ensure computational robustness and cross-modality generalization, we utilize two fundamentally distinct datasets: the UCSF DANDI dataset (high-density surface ECoG capturing English syllables) and the Maastricht *SingleWordProductionDutch-iBIDS* dataset (sparse, deep-brain sEEG capturing Dutch words).

To model the highly non-linear relationship between neural activity and acoustic features, we engineered a two-stage machine learning pipeline. First, broadband high-gamma frequency envelopes were extracted and synchronized with Text-to-Speech (TTS) generated Mel-spectrogram targets. Second, a baseline Linear Regression model was employed as a signal integrity gate, followed by a generative Diffusion model. By iteratively denoising spectrogram predictions explicitly guided by the extracted neural features, the diffusion architecture successfully synthesizes complex acoustic structures that linear models fail to resolve.

Experimental results demonstrate that the diffusion model achieves strong temporal alignment across both modalities, yielding peak Pearson correlations of 0.97 on the sEEG dataset and 0.91 on the ECoG dataset. However, qualitative evaluation revealed a critical discrepancy between statistical metrics and perceptual audio quality. While the deep-brain sEEG modality captured sufficient spectral variance to produce intelligible audio, the results demonstrate that diffusion mapping of surface-level ECoG kinematics—while achieving highly accurate temporal boundaries—produces magnitude-only spectrogram predictions that are too rigid to synthesize internal frequency gradients. Consequently, the ECoG waveforms suffered from destructive interference and severe amplitude collapse during Griffin-Lim phase reconstruction. This work provides a mathematically robust, systematic framework for neural speech decoding, framing the phase-retrieval bottleneck as a critical scientific finding that mandates the future integration of advanced neural vocoders.

Chapter 1

Introduction

This thesis explores the development of advanced generative models capable of reconstructing spoken audio directly from intracranial neural recordings. By leveraging modern deep learning architectures—particularly diffusion-based generative models—we aim to create neural speech synthesis solutions that are highly accurate, robust across different brain morphologies, and clinically practical.

1.1 Context and Motivation

Human communication is fundamentally rooted in the ability to produce fluid, intelligible speech. However, a variety of severe neurological conditions—including Amyotrophic Lateral Sclerosis (ALS), brainstem strokes, and high spinal cord injuries, can entirely sever the neural pathways that control the vocal tract. This leads to anarthria (the complete loss of speech articulation), often culminating in locked-in syndrome. In this state, patients retain full cognitive function and awareness but are trapped within paralyzed bodies, stripped of their primary means of interacting with the world [20].

Historically, assistive communication technologies have relied on residual voluntary muscle movements, such as eye-tracking or subtle cheek twitches. While these devices provide a vital lifeline, they are fundamentally limited by their bandwidth, typically allowing users to produce only 10 to 15 words per minute. This stands in stark contrast to the 150 to 200 words per minute characteristic of natural conversational speech [15]. Furthermore, spelling interfaces strip communication of its acoustic richness, removing the emotional prosody, tone, and identity inherent in a human voice.

Over the past decade, Brain-Computer Interfaces (BCIs) have emerged as a transformative solution to this communication bottleneck. Rather than relying on intact peripheral muscles, speech neuroprosthetics bypass damaged pathways entirely. By recording high-resolution neural activity directly from the cortical surface specifically targeting the ventral sensorimotor cortex (vSMC), which orchestrates the complex kinematics of the lips, jaw, tongue, and larynx [3]—BCIs can intercept the brain’s raw intention to speak. The ultimate goal of speech BCI research is to translate these intercepted neural signals not just into text on a screen, but directly into synthesized, natural-sounding audio in real-time [16].

1.2 Problem Statement

Translating neural activity directly into intelligible audio is a formidable computational challenge. While intracranial recordings provide the highest fidelity windows into cortical function, utilizing them for speech synthesis introduces three overlapping data and modeling hurdles:

1. **Inherent Noise and High Dimensionality:** Brain signals possess a notoriously low signal-to-noise ratio. Intracranial arrays record continuous neural activity where target speech-motor control is superimposed over myriad background cognitive processes, autonomic functions, and environmental electrical artifacts. While these concurrent neural processes are biologically meaningful, they act as irrelevant noise within the specific context of a speech decoding task. Furthermore, the data is highly dimensional; a standard ECoG array produces a continuous stream of up to 256 parallel channels sampled thousands of times per second, requiring highly specialized spatiotemporal architectures to isolate relevant articulatory features.
2. **Indirect Biological Mapping:** The ventral sensorimotor cortex (vSMC) primarily encodes articulatory motor commands (the biomechanical kinematics required to move physical articulators) rather than direct acoustic output. While regions such as the superior temporal gyrus (STG) contain representations more closely related to auditory feedback, the primary drive signals from the vSMC require translation. Consequently, there is no simple one-to-one or linear mathematical mapping between a recorded neural voltage and a target audio waveform. The model must learn the deeply complex, non-linear inference connecting a motor plan to its resulting acoustic consequence.
3. **Extreme Inter-Patient Variability:** A critical bottleneck in modern speech neuroprosthetics is the difficulty of cross-subject generalization. Human cortical anatomy is highly idiosyncratic, and electrode placement is dictated entirely by individual clinical necessity rather than standardized recording grids. Because spatial topographies differ drastically between subjects, a neural decoder trained on one participant typically exhibits substantial performance degradation when applied directly to another without extensive adaptation. Overcoming this patient-specific barrier remains one of the central challenges in deploying scalable BCI technologies.

1.3 The Technical Challenge: iEEG and Speech Decoding

Translating the intent to speak into intelligible audio requires capturing brain signals with immense precision. Recent progress in neural signal acquisition and machine learning has paved the way for decoding speech-related brain activity directly from the cortex.

Among various neuroimaging modalities, intracranial electroencephalography (iEEG) offers a unique advantage. By placing electrodes directly inside the brain volume or on the cortical surface, iEEG bypasses the signal degradation of the skull, providing the exceptionally high temporal and spatial resolution necessary to capture the rapid neural dynamics underlying continuous speech. Previous benchmark studies have successfully demonstrated the feasibility of decoding discrete phonemes, words, and even text sentences from this cortical activity [21]. However, advancing from text-generation to the direct synthesis of natural-sounding, continuous audio remains a formidable computational challenge.

1.4 Proposed Approach and Methodology

To address this synthesis bottleneck, this project leverages two fundamentally distinct, publicly available neurophysiological datasets. The first, provided by Verwoert et al. [21], captures deep-brain stereotactic EEG (sEEG) recordings during continuous, whole-word production. The second, curated by Dr. Edward Chang’s laboratory at UCSF [3], provides high-density surface Electrocorticography (ECoG) recordings isolating the precise kinematics of consonant-vowel syllables. Together, these datasets provide a comprehensive foundation for training and cross-validating deep learning models across differing brain morphologies, hardware modalities, and linguistic paradigms.

We specifically focus on mapping extracted high-gamma neural features to their corresponding Mel-spectrogram representations, aiming to directly reconstruct the underlying acoustic signal of the human voice. To achieve this, we developed a two-stage neural network architecture. The pipeline utilizes an initial feedforward network to generate a coarse structural estimate of the audio, which is subsequently passed to an advanced diffusion model to iteratively refine the acoustic textures.

Rather than attempting to map ambiguous brain signals to pristine audio in a single, error-prone step, the diffusion framework utilizes a controlled probabilistic process. During training, statistical Gaussian noise is progressively added to the ground-truth spectrograms until the acoustic data is completely corrupted. The deep learning model is then trained to reverse this exact process, mathematically estimating and subtracting the noise step-by-step to recover the clean audio matrix, guided entirely by the neural features.

This counter-intuitive mechanism—destroying data to learn how to rebuild it—is highly advantageous for speech neuroprosthetics. By learning the fundamental "rules" and structural manifold of valid human speech during the denoising process, the diffusion model can synthesize sharp, high-fidelity acoustic textures even when the input neural signals are highly ambiguous, effectively overcoming the oversmoothed, blurry outputs that plague traditional regression models.

1.5 Significance and Ultimate Goal

The methodology adopted in this thesis builds directly upon recent advances in deep generative learning for biomedical signal processing. The ultimate goal of this research is to contribute to the rapidly growing field of neural speech decoding by rigorously evaluating how accurately localized intracranial signals can predict high-fidelity speech acoustics across diverse biological hardware.

Unlike traditional Automatic Speech Recognition (ASR) systems that rely on physical acoustic input, this approach taps directly into the brain’s internal kinematic representation of speech. By benchmarking the performance of a diffusion-based framework across multiple highly complex, real-world datasets—spanning both deep-brain penetrating shafts (sEEG) and high-density surface arrays (ECoG)—this research maps the performance bounds and generalization capabilities of modern generative neuroprosthetics.

Ultimately, this work offers a viable computational path toward continuous, naturalistic communication restoration for completely locked-in patients [16, 21]. We believe that engineering efforts of this nature are vital for translating abstract neuroscientific insights into practical, life-altering assistive technologies.

1.6 Objectives and Contributions

The primary objective of this thesis is to develop a robust, end-to-end computational pipeline capable of synthesizing intelligible speech directly from human intracranial recordings across differing recording modalities. This research makes the following specific contributions:

- **Neural Feature Extraction Pipeline:** Designed and implemented a signal processing framework to extract broadband high-gamma (70 Hz – 170 Hz) envelopes from raw ECoG and sEEG recordings, isolating the frequencies most highly correlated with speech motor kinematics.
- **Acoustic Target Alignment and TTS Augmentation:** Developed a temporal alignment protocol to synchronize neural activity with speech events. To overcome dataset anonymity limitations, an automated spatial-parsing algorithm was engineered to infer patient profiles, enabling the generation of clean, gender-matched Mel-spectrogram ground-truth targets using a Text-to-Speech (TTS) engine.
- **Two-Stage Machine Learning Architecture:** Engineered a hybrid machine learning model utilizing a Linear Regression (LR) model as an initial baseline gating mechanism to verify signal integrity, followed by an advanced neural diffusion refiner designed to learn the non-linear mapping to high-resolution Mel-spectrograms.
- **Cross-Modality Validation:** Demonstrated the robustness of the proposed architecture by deploying it across two fundamentally distinct open-access neurophysiological datasets: dense surface ECoG grids (English syllables) and sparse deep-brain sEEG shafts (Dutch whole words).

1.6.1 Scientific Contributions and Recognition

This work involved the independent design, implementation, and evaluation of the entire diffusion model pipeline. The underlying neural recordings and associated metadata were obtained from publicly available repositories [3, 21]. The resulting research was presented at the NVHH Conference (Hungary) and the Beszédkutatás – Speech Research Conference [1], and received an award at the local TDK competition.

1.7 Thesis Roadmap

The remainder of this thesis is organized as follows:

Chapter 2 establishes the theoretical foundations of the study, detailing the neuroscience of speech motor control, the hardware distinctions between intracranial recording modalities, the mechanics of generative diffusion models, and the current state-of-the-art in speech neuroprosthetics.

Chapter 3 provides a comprehensive technical breakdown of the two distinct neurophysiological datasets utilized in this research: the Maastricht Dutch Words (sEEG) dataset and the UCSF English Syllables (ECoG) dataset.

Chapter 4 outlines the proposed end-to-end computational architecture, encompassing raw data ingestion, high-gamma feature extraction, dimensionality reduction, the two-

stage diffusion-based generative network, and final waveform synthesis via the Griffin-Lim algorithm.

Chapter 5 presents the quantitative and qualitative experimental results. It directly compares the foundational linear regression baseline against the generative diffusion pipeline across both recording modalities, concluding with a rigorous physiological and digital signal processing analysis of waveform amplitude collapse.

Chapter 6 summarizes the core findings of this research, formally defines the biological and computational limitations of the study, and outlines critical avenues for future work required to achieve clinical viability.

Finally, the document concludes with the author’s academic publications, mandatory declarations, and the bibliography.

Chapter 2

Background and Related Work

This chapter provides the theoretical and empirical foundations for this study, bridging the neurobiological mechanisms of human speech with the generative machine learning frameworks used to decode them.

2.1 The Neuroscience of Speech Motor Control

In computational terms, human speech production can be modeled as a highly complex, closed-loop control system. The execution of speech is distributed across a network of specialized cortical processors, predominantly mediated by the ventral sensorimotor cortex (vSMC), the inferior frontal gyrus (IFG), and the superior temporal gyrus (STG). Within this network, the IFG is believed to contribute to higher-level speech planning and sequencing. The vSMC functions as the primary motor controller, dispatching millisecond-scale motor commands to the physical articulators (e.g., respiratory, laryngeal, and facial muscles). Concurrently, the STG processes auditory feedback, serving as the error-correction mechanism in the control loop [7, 3].

A fundamental challenge in training machine learning models to decode this process lies in understanding the brain’s internal feature representation. Foundational work utilizing high-density electrocorticography (ECoG) arrays provided strong evidence that the vSMC exhibits somatotopic organization [3]. This indicates that neighboring neural populations tend to represent neighboring articulatory functions. The neural populations cluster functionally around *kinematic* states (i.e., the mechanical gestures required to close the lips or elevate the tongue) rather than mapping directly to the output acoustic frequency domain. Consequently, the task of any downstream deep learning model is to compute a highly non-linear translation from a mechanical actuator state-space to a target audio spectrogram.

From a signal processing perspective, extracting actionable features from raw cortical voltage traces requires isolating specific frequency bands. Researchers focus almost exclusively on the broadband *high-gamma* frequency range (70 Hz – 170 Hz). Unlike lower frequency bands (e.g., alpha or beta waves), which represent broad, synchronous, and largely oscillatory resting states, high-gamma power is non-oscillatory. It is strongly correlated with local population firing activity [19]. In other words, extracting the high-gamma envelope effectively isolates the specific activation spikes of the local neural clusters directly beneath the electrode, filtering out global background noise.

Furthermore, recent advancements leveraging deep-brain stereotactic EEG (sEEG) have expanded our mapping of the speech network beyond the 2D cortical surface. By using penetrating depth probes, researchers can capture the signal dynamics of sub-cortical structures such as the hippocampus and thalamus. These structures are believed to contribute to memory, cognitive control, and temporal coordination processes that influence speech production, providing critical temporal context for decoding algorithms [21].

2.1.1 Somatotopic Representation of Phonetic Features

To fully appreciate the complexity of training generative models on neural data, one must understand the precise somatotopic organization of the ventral sensorimotor cortex (vSMC). The dataset utilized in this study, initially mapped by Bouchard et al. [3], demonstrates that within the vSMC, neural activity is organized primarily around articulatory movements rather than acoustic output. It predominantly reflects the physical location and degree of vocal tract constriction. The vSMC is organized along a dorsal-to-ventral axis, structurally mirroring the physical articulators: dorsal regions control the lips, mid-regions control the tongue tip and jaw, and ventral regions control the posterior tongue body and larynx.

However, the neural control of speech is not as simple as activating a single "lip" or "tongue" electrode. The vSMC dynamically encodes highly nuanced phonetic contrasts through distinct spatiotemporal activation patterns. These kinematics can be categorized by their specific mechanical requirements:

Constriction Location (Fricatives): Fricative consonants are produced by forcing air through a narrow channel in the vocal tract, creating turbulent, hissing acoustics. For example, producing the unvoiced fricatives $/\theta/$ (as in "thin"), $/s/$ (as in "sin"), and $/f/$ (as in "shin") all require the tongue tip. Yet, the physical location of the constriction moves progressively backward from the teeth ($/\theta/$), to the alveolar ridge ($/s/$), to the hard palate ($/f/$). The vSMC distinguishes these sounds by shifting the epicenter of high-gamma activation across distinct, millimeter-scale neural populations within the mid-vSMC tongue representation. A successful neural decoder must be sensitive enough to detect these microscopic spatial shifts to synthesize the correct acoustic friction.

Constriction Degree and Shape (Coronals): Beyond simply moving an articulator to a specific location, the brain must dictate how tightly that articulator presses against the vocal tract. Consider the front-tongue consonants $/l/$, $/n/$, and $/d/$. All three require the tongue tip to contact the alveolar ridge behind the top teeth. However, $/d/$ requires a complete, forceful stoppage of air (a plosive), $/n/$ requires complete oral closure while dropping the velum to allow air through the nose (a nasal), and $/l/$ requires the tongue tip to touch the ridge while allowing air to flow over the sides of the tongue (a lateral approximant). The neural signatures for these three sounds overlap heavily in spatial location but differ drastically in their temporal activation intensity and duration.

Coarticulation and Vowel Context: Perhaps the most complex hurdle for machine learning models is the biophysical phenomenon of coarticulation. Human speech is not a sequence of isolated, robotic sounds; it is fluid. The neural motor plan for a specific consonant changes dynamically based on the vowel that follows it. For instance, when a speaker produces the consonant $/j/$ (the "y" sound in "yes") followed by the vowel $/i/$ (as in "yield"), the vSMC shows sustained, prolonged activation in the tongue electrodes, as the tongue must remain elevated for the high-front vowel. Conversely, if the same $/j/$ is followed by the vowel $/u/$ (as in "youth"), an entirely different neural population in the dorsal vSMC activates simultaneously to prepare the lips for rounding [3].

This biological reality—where the neural signature of a single consonant changes depending on its surrounding context—provides a strong physiological motivation for the feature engineering choices made in this thesis. It supports the temporal "Feature Stacking" (incorporating ± 4 preceding and succeeding temporal windows) outlined in the methodology, ensuring the generative model has access to the neural anticipation and lingering motor dynamics inherent in natural human speech.

2.2 Intracranial Recording Modalities

Brain-Computer Interfaces (BCIs) fundamentally rely on the acquisition of high-fidelity neural signals. While non-invasive modalities like scalp EEG are highly accessible, they suffer from severe spatial blurring and signal attenuation due to the insulating properties of the skull and cerebrospinal fluid. This biological low-pass filtering effectively destroys the high-frequency micro-textures (such as the high-gamma band) required for complex speech decoding. Consequently, state-of-the-art speech neuroprosthetics rely exclusively on intracranial electroencephalography (iEEG), which places recording hardware directly inside the cranial cavity. The quality, spatial distribution, and nature of the captured data are dictated entirely by the chosen hardware modality: Electrocorticography (ECoG) or Stereotactic EEG (sEEG) [17].

2.2.1 Electrocorticography (ECoG)

ECoG utilizes flexible, multi-contact silastic grids or strips placed directly on the exposed subdural or epidural surface of the cortex. The implantation of these arrays requires a surgical craniotomy (the temporary removal of a section of the skull).

Strengths: Because the electrode grid sits directly on top of the brain tissue like a blanket, ECoG provides a dense, continuous two-dimensional map of cortical surface activity. It offers exceptional spatial resolution (often at a 3–4 mm pitch) and an incredibly high signal-to-noise ratio in the high-gamma frequency band (70–170 Hz) [9]. Because the primary speech motor networks (the ventral sensorimotor cortex) and auditory processing centers (the superior temporal gyrus) are located predominantly on the lateral surface of the brain, ECoG has historically served as the gold standard for speech BCI research [3].

Limitations: The primary biological limitation of ECoG is its strict confinement to the cortical surface. The human brain is highly convoluted; approximately two-thirds of cortical tissue resides hidden within deep folds (sulci). ECoG cannot record from these sulcal depths, nor can it access critical sub-cortical memory and timing structures. Clinically, the requirement of a large craniotomy introduces significant surgical morbidity risks, making ECoG less ideal for permanent, lifetime BCI implants.

2.2.2 Stereotactic EEG (sEEG)

sEEG represents a fundamentally different surgical and electrophysiological approach. Rather than resting on the surface, sEEG utilizes rigid, penetrating electrode shafts (often containing 10 to 15 individual cylindrical contacts) inserted deep into the brain volume through tiny, stereotactically guided burr holes in the skull [13].

Strengths: The defining advantage of sEEG is its volumetric reach. It uniquely captures the neural dynamics of deep sulcal folds, the insula, the hippocampus, and the basal ganglia—regions deeply involved in the cognitive, lexical, and temporal planning of speech

[21]. Furthermore, because it avoids a massive craniotomy, sEEG carries a significantly lower risk of surgical complications (such as hemorrhage or infection). The surrounding brain tissue acts as an anchor for the rigid shafts, providing highly stable electrical impedance over time. This makes sEEG the leading candidate for chronic, long-term BCI implants designed for daily, at-home use.

Limitations: The trade-off for this deep-brain access is spatial sparsity. An sEEG shaft effectively samples a "tunnel" of tissue. While it provides excellent depth resolution along the shaft's trajectory, it leaves massive spatial blind spots between the shafts. Unlike ECoG, sEEG cannot capture a contiguous, high-density 2D topographic map of an entire motor region.

2.2.3 Comparative Impact on Machine Learning

From a computational modeling perspective, developing a generalized machine learning architecture capable of handling both modalities is a formidable challenge.

Table 2.1: Comparative analysis of intracranial recording modalities and their implications for BCI decoding.

Feature	ECoG (Surface Grids)	sEEG (Penetrating Shafts)
Spatial Coverage	Continuous 2D surface map	Sparse 3D volumetric sampling
Depth Access	Surface gyri only (misses sulci)	Accesses deep sulci and sub-cortex
Clinical Invasiveness	High (requires large craniotomy)	Low (requires tiny burr holes)
Implant Stability	Prone to micro-shifting over time	Highly stable for chronic implants
Decoding Advantage	Captures dense, contiguous kinematic topographies	Captures deep cognitive/temporal context

An architecture trained on ECoG (such as the UCSF dataset utilized in this study) learns from highly localized, dense kinematic topographies. Conversely, an architecture trained on sEEG (such as the Maastricht dataset) must learn to synthesize speech from a structurally sparse but cognitively deep feature space. Demonstrating that the proposed Conditional Diffusion pipeline can successfully synthesize intelligible Mel-spectrograms across both of these vastly different biological hardware paradigms is a primary objective of this thesis.

2.3 Generative AI and Diffusion Models

The current vanguard of high-fidelity acoustic synthesis relies on generative deep learning frameworks. Unlike traditional regression models—which tend to average their outputs and produce blurry, muffled audio—generative models are trained to learn the underlying structural rules of a dataset, allowing them to synthesize sharp, highly realistic spectral representations.

To overcome the training instability and latency issues of older generative models (such as GANs), the state-of-the-art has shifted toward Denoising Diffusion Probabilistic Models (DDPM) [10]. Diffusion models operate by conceptualizing generation as a two-step process of controlled destruction and reconstruction:

- **Forward Process (Adding Noise):** The model takes a ground-truth Mel-spectrogram and iteratively adds increments of Gaussian noise over hundreds of timesteps, eventually reducing the acoustic features to unstructured Gaussian static.
- **Reverse Process (Denoising):** A deep neural network (typically a U-Net) is trained to reverse this process. By learning to incrementally predict and subtract the noise at each step, the network learns the fundamental spectrotemporal structures of human speech, allowing it to reconstruct a coherent spectrogram from pure randomness.

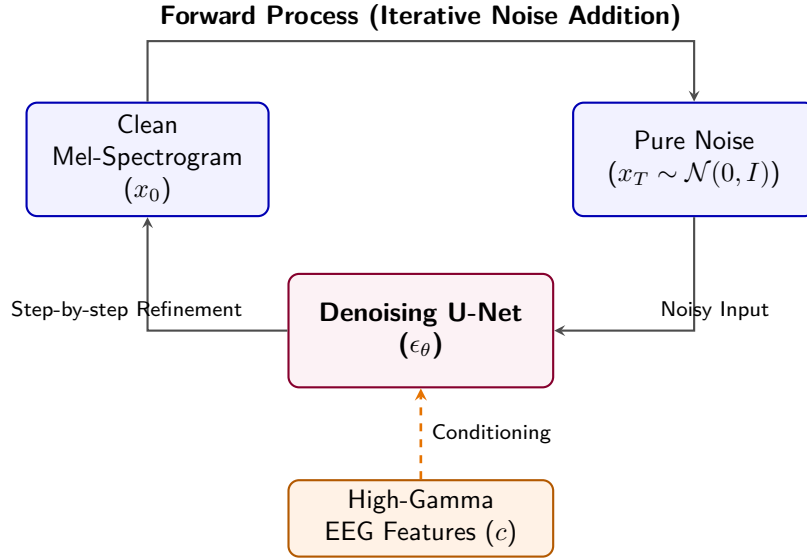


Figure 2.1: Conceptual overview of the diffusion process. The forward process degrades the target Mel-spectrogram into pure Gaussian noise. The reverse process employs a U-Net to iteratively denoise the matrix. In this thesis, the denoising trajectory is explicitly conditioned on the patient’s high-gamma neural features to synthesize the intended speech.

By default, a standard diffusion model generates random variations of the data it was trained on. However, for a speech neuroprosthetic, the network must generate the specific utterance intended by the patient. To achieve this, the reverse denoising process is explicitly *conditioned* on the high-gamma neural representations. As the network iteratively removes noise, these cortical features act as a biological blueprint, guiding the reconstruction to sculpt the correct vocal formants and consonant bursts. This probabilistic approach overcomes the oversmoothed outputs typical of regression baselines, yielding the rich, high-frequency spectral textures required for naturalistic speech.

2.4 Acoustic Representation: The Target Domain

While the input space of a speech neuroprosthetic consists of high-dimensional neural voltages, the target output space must be carefully engineered to represent human speech efficiently. Raw audio waveforms are one-dimensional time-series sampled at highly dense rates (e.g., 16,000 to 48,000 samples per second). Training deep generative models to predict audio sample-by-sample is computationally exorbitant and mathematically unstable, as the model must learn both the global semantic structure and the microscopic temporal oscillations simultaneously.

To bypass this bottleneck, speech synthesis architectures overwhelmingly target intermediate time-frequency representations—specifically, the logarithmic Mel-spectrogram. By applying the Short-Time Fourier Transform (STFT), the 1D audio waveform is decomposed into a 2D matrix representing the magnitude of specific frequency bands over discrete temporal windows.

Crucially, human auditory perception is not linear. The cochlea is highly sensitive to microscopic changes in low frequencies (where fundamental pitch and vowel formants reside) but increasingly insensitive to changes in higher frequencies (where fricative noise resides). To mimic this biological reality, the linear frequency bins of the STFT are mapped onto the Mel scale, a non-linear perceptual scale where equal distances in pitch sound equally distant to the human ear. This mapping applies a logarithmic transformation to the linear frequencies, compressing them into distinct psychoacoustic bins that mirror the resolution of the human auditory system.

By predicting Mel-spectrograms rather than raw audio, the deep learning model is forced to optimize for the acoustic features most critical to human intelligibility, drastically reducing the dimensionality of the target space while preserving the structural acoustic envelope.

2.5 The Phase Retrieval Bottleneck

The mathematical conversion of a waveform into a spectrogram introduces a fundamental digital signal processing limitation: the loss of phase. A complex spectrogram generated by an STFT contains both magnitude (the amplitude of the frequencies) and phase (the temporal alignment of the sine waves).

Deep learning models excel at predicting magnitude because it contains visible, structural patterns (e.g., the dark bands of vocal formants). Phase, conversely, is notoriously chaotic, wrapping rapidly between $-\pi$ and π without a predictable spatial structure. Consequently, generative models—including the diffusion architectures utilized in this thesis—are trained exclusively to predict the magnitude Mel-spectrogram, discarding the phase entirely.

To synthesize listenable audio, the missing phase must be artificially estimated. Historically, this has been achieved using the Griffin-Lim Algorithm (GLA) [6], which iteratively projects the magnitude spectrogram back and forth between the time and frequency domains to reconstruct a phase estimate that is consistent with the observed magnitudes. While modern neural vocoders (such as WaveNet or HiFi-GAN) can generate highly realistic phase data, they introduce secondary black-box neural networks into the pipeline. In neuroprosthetic research, utilizing classical algorithms like Griffin-Lim is often preferred during model evaluation, as it ensures that the perceived audio quality reflects the raw predictive power of the primary neural decoder, rather than the smoothing artifacts of an external vocoder.

2.6 Standardized Evaluation Metrics

Comparing the efficacy of differing deep learning architectures requires rigorous, standardized quantitative metrics. Because the primary objective of the proposed architecture is to synthesize the structured acoustic envelope from neural data, the quantitative analysis in this thesis relies on the Pearson Correlation Coefficient (r).

The Pearson Correlation Coefficient is the most ubiquitous metric in speech Brain-Computer Interface (BCI) literature. It measures the linear temporal correlation between the predicted Mel-spectrogram magnitude and the ground-truth target spectrogram. In the context of speech neuroprosthetics, it evaluates the structural and temporal accuracy of the generated acoustic features, with a score of 1.0 indicating a perfect prediction.

By calculating the correlation of the generated spectrogram magnitudes directly against the targets, Pearson’s r effectively isolates the neural network’s core translation capabilities. This provides a robust, objective measure of the generative model’s performance without relying on subjective human-listening scores or downstream audio post-processing.

2.7 Literature Review and State of the Art

Speech neuroprosthetics have evolved from abstract classifications toward continuous, personalized audio synthesis. This section reviews the core literature and recent models relevant to the experiments in this thesis.

2.7.1 Linear Regression Baselines: The Benchmark

A key milestone for reproducible neuroprosthetic research was the open-access release of the *SingleWordProductionDutch-iBIDS* dataset by Verwoert et al. [21]. In their initial technical validation, the authors demonstrated that a basic Ordinary Least Squares (OLS) linear regression framework—mapping deep sEEG signals directly to log-Mel spectrograms—could successfully capture the macro-structure of the audio. Specifically, the linear model was highly effective at differentiating active, continuous speech intervals from baseline silence.

However, from a machine learning perspective, direct linear mappings are fundamentally insufficient for high-fidelity acoustic synthesis. Linear regression assumes a strictly proportional, additive relationship between localized high-gamma neural bursts and the resulting acoustic frequencies. Human speech, conversely, is highly non-linear; the physical articulators (tongue, lips, jaw) interact dynamically to produce complex harmonic textures, turbulent noise bursts, and rapid spectral transitions.

When a linear weight matrix attempts to map these complex dynamics, it inevitably suffers from severe *oversmoothing*. The model optimizes for the lowest mean squared error by predicting the statistical "mean" of the spectrogram. While this successfully captures the overall amplitude envelope (the gross timing and loudness), it completely blurs the sharp, distinct frequency bands required for intelligible speech. Consequently, when these oversmoothed spectrograms are passed to a phase-reconstruction algorithm (such as Griffin-Lim), the resulting audio is heavily degraded, lacking the crisp acoustic boundaries of natural human speech.

Rather than discarding this baseline, this thesis strategically repurposes the linear framework. Because linear regression is deterministic, highly computationally efficient, and re-

liable at capturing coarse temporal boundaries, it serves as an ideal Stage-1 "gatekeeper." In the proposed two-stage pipeline, the linear baseline is first trained on the PCA-reduced neural features to verify basic signal integrity and temporal alignment. Only if this linear baseline achieves a minimum correlation threshold (demonstrating that the neural sequence contains valid speech kinematics) is the session passed to the computationally expensive Stage 2: the generative Conditional Diffusion Refiner. This approach formally bridges the gap between the established linear benchmarks of the past and the non-linear, generative architectures required for the future of speech neuroprosthetics.

2.7.2 Personalized BCIs: Zero-Shot Voice Conversion from Neural Data

Historically, speech neuroprosthetics have focused entirely on decoding semantic content determining *what* the patient is trying to say. However, recent literature has expanded toward restoring a patient’s vocal identity, focusing on *how* they sound. Standard speech synthesizers often rely on generic, robotic voice profiles, which strips patients of their unique biometric identity (e.g., their specific pitch, tone, and vocal timbre).

To address this, Geng et al. [5] introduced a "zero-shot" voice conversion framework driven entirely by target EEG recordings. Building upon prior advancements in neural speaker recognition [23, 11] and imagined speech reconstruction [18, 22], their architecture leverages cross-modal latent space alignment. Specifically, the system maps neural embeddings (extracted directly from brain waves) to the acoustic embeddings of a pretrained, speech-only generative model.

Remarkably, when evaluated on the *SingleWordProductionDutch* dataset, the researchers demonstrated that neural activity inherently contains sufficient structural and biometric information to approximate an individual’s unique vocal timbre. Because the framework is "zero-shot," it is capable of synthesizing a personalized voice profile purely from neural kinematics, without requiring any prior acoustic training data of the target speaker’s actual voice. This represents a critical step forward for clinical BCIs, offering a pathway to fully restore vocal identity for patients suffering from locked-in syndrome or severe dysarthria.

2.7.3 Transformer Architectures: The MiSTR Framework

As speech neuroprosthetics have evolved, researchers recognized that traditional convolutional and recurrent architectures (CNNs and RNNs) struggle to effectively capture the long-range temporal dependencies inherent in continuous human speech. Human articulation is heavily influenced by *coarticulation*, the biophysical phenomenon where the vocal tract prepares for upcoming phonemes while still executing the movements of current ones. To address this sequential bottleneck, Al-Radhi et al. [2] introduced the MiSTR (Multi-Modal iEEG-to-Speech Synthesis) framework, representing a pivotal shift toward attention-based sequence modeling in brain-computer interfaces.

The MiSTR architecture systematically replaces classical recurrent layers with a highly parallelized Transformer-based decoder. Computationally, Transformers utilize *self-attention mechanisms*, which allow the network to simultaneously evaluate the entire temporal sequence of neural activity. Instead of processing brain waves rigidly frame-by-frame, the self-attention matrix dynamically weights the relevance of distant neural contexts looking both forward and backward in time to accurately predict the corresponding acoustic target at any given millisecond.

Furthermore, the MiSTR framework innovates on the feature extraction front by replacing standard Short-Time Fourier Transforms (STFT) with a wavelet-based neural encoder. Intracranial signals are highly non-stationary; transient consonant bursts require high temporal resolution, while sustained vowel phonation requires high spectral resolution. By utilizing multi-resolution continuous wavelet transforms, the encoder captures a significantly richer, physiological time-frequency representation without falling victim to the fixed-window limitations of STFT.

Crucially, Al-Radhi et al. recognized that intelligible, natural-sounding speech requires more than just phonetic content. The MiSTR framework jointly models traditional high-gamma articulatory gestures alongside complex cross-frequency interactions (such as how low-frequency neural rhythms modulate high-frequency bursts). It also explicitly decodes prosodic features, specifically fundamental frequency (pitch) and acoustic intensity (loudness). By grounding the Transformer’s generative output in both phonetic structure and human prosody, the MiSTR framework achieved an unprecedented state-of-the-art mean Pearson correlation of 0.91 on public iEEG benchmarks. This milestone mathematically established that attention-based architectures can effectively unravel the deeply non-linear relationship between cortical kinematics and high-fidelity continuous speech.

2.7.4 Diffusion Models in Acoustic Synthesis

Recently, Denoising Diffusion Probabilistic Models (DDPMs) have emerged as the dominant framework for generative synthesis. In the broader field of speech technology, diffusion-based "neural vocoders"—such as DiffWave [14] and FastDiff [12]—revolutionized synthesis by generating raw audio waveforms directly, bypassing classical signal-processing vocoders.

However, the pipeline proposed in this thesis adopts a distinct approach: instead of attempting raw waveform synthesis (which necessitates prohibitively high sampling rates and computational costs for continuous intracranial decoding), this architecture utilizes diffusion to synthesize logarithmic Mel-spectrograms. This "spectrogram-first" strategy provides a robust intermediate representation that captures the necessary acoustic energy while maintaining a computationally tractable latent space for iterative denoising.

Following breakthroughs in standard Text-to-Speech (TTS), researchers have begun adapting these mechanisms for Brain-Computer Interfaces. Qi et al. [18] introduced an AudioDiffusion framework conditioned on non-invasive scalp EEG signals. However, the application of conditional diffusion models to highly localized, invasive intracranial recordings (such as ECoG and sEEG) for continuous speech decoding remains at the absolute cutting edge of the field. This thesis addresses this frontier by uniquely conditioning the reverse denoising process on high-gamma cortical kinematics to reconstruct naturalistic Mel-spectrogram representations, which are subsequently inverted into audio via phase retrieval.

2.8 Summary and Thesis Context

The established trajectory of the literature in speech neuroprosthetics demonstrates a clear, rapid evolutionary path. Early foundational works relied heavily on basic linear regression to establish benchmark mappings between cortical activity and crude acoustic envelopes. As the field matured, the integration of convolutional and recurrent neural networks (CNNs and RNNs) allowed researchers to capture localized spatiotemporal con-

text, advancing the field from mere speech detection to the intelligible reconstruction of phonetic features. More recently, the paradigm has shifted toward attention-based architectures, with Transformers successfully resolving the long-range temporal dependencies and coarticulation challenges inherent in continuous speech. Finally, the state-of-the-art has embraced generative diffusion models to synthesize the highly detailed, non-linear acoustic textures required to restore naturalistic human timbre.

However, despite these computational leaps, significant challenges remain regarding cross-modality generalization and architectural efficiency. Many advanced deep learning models in the current literature are highly optimized for specific hardware types or single-patient cohorts, leaving their adaptability across diverse neurophysiological recording environments largely unexplored.

This thesis directly addresses these critical gaps by systematically building upon the established evolutionary paradigm. It explicitly evaluates and compares two distinct tiers of computational complexity: a foundational Linear Regression baseline (serving as a temporal gatekeeper and benchmark), and a state-of-the-art Conditional Diffusion model (to iteratively synthesize high-fidelity Mel-spectrograms). By rigorously testing these two architectures across fundamentally distinct recording modalities deep-brain penetrating sEEG (Dutch whole-word production) and dense surface-level ECoG (English consonant-vowel syllables) this research maps out the respective performance bounds, computational trade-offs, and cross-modality generalization capabilities of modern generative speech neuroprosthetics.

Chapter 3

Datasets

Translating raw neural activity directly into audible speech—often termed neural speech synthesis—represents one of the most complex frontiers in Brain-Computer Interface (BCI) research. The natural production of speech requires a highly synchronized, sub-second orchestration of cognitive planning and motor execution. Because this capacity is uniquely human, its neural underpinnings cannot be modeled in animals; researchers must rely exclusively on high-fidelity, invasive recordings to map these networks [21]. However, because such data can only be ethically obtained during specialized clinical procedures, the scarcity of accessible human recordings remains a primary bottleneck in developing robust brain-to-speech systems.

To navigate this data scarcity and ensure the computational robustness and cross-modality generalization of the proposed neural speech synthesis pipeline, this study utilizes two fundamentally distinct, publicly available neurophysiological datasets. The first leverages deep-brain stereotactic Electroencephalography (sEEG) to capture Dutch whole-word production, while the second utilizes dense, surface-level Electrocorticography (ECoG) to map English consonant-vowel syllables.

3.1 Dataset 1: Maastricht Dutch Words (sEEG)

The first dataset utilized in this study is the **SingleWordProductionDutch-iBIDS** corpus [21], published in *Nature Scientific Data*. Collected collaboratively by Maastricht University and the Epilepsy Center Kempenhaeghe, this dataset provides a unique multi-modal pairing of neural and acoustic information. It comprises high-temporal-resolution intracranial recordings from 10 participants, capturing neural activity across 1,103 total electrodes while the subjects read individual words aloud.

Crucially, this extensive coverage of both cortical and sub-cortical topographies provides an ideal testbed for advancing speech neuroprostheses. By making these high-fidelity recordings publicly available, the authors aimed to accelerate the exact type of research conducted in this thesis: testing and validating novel computational approaches for decoding and synthesizing speech directly from neural timeseries.

While the majority of speech Brain-Computer Interface (BCI) research has historically relied on surface-level ECoG grids, this dataset employs sEEG. sEEG involves implanting electrode shafts deep into the brain volume, providing sparse but widespread sampling of both cortical areas and deep sub-cortical structures. This is highly advantageous for modern speech synthesis models; recent neurophysiological advances demonstrate that

speech production is not limited to the surface motor cortex, but heavily relies on deep structures such as the hippocampus and basal ganglia for sequence planning and lexical retrieval. By utilizing sEEG, this dataset provides the deep-brain kinematic data necessary to model the complete speech production network.

3.1.1 Subjects and Clinical Paradigm

The corpus comprises neural and acoustic recordings from 10 individuals (five male, five female; mean age = 32 years; range = 16–50 years). All participants were native Dutch speakers with no documented cognitive, linguistic, or speech-motor deficits, ensuring that the recorded neural kinematics represent normative speech production networks.

All participants in this cohort were diagnosed with severe, pharmacoresistant epilepsy. Because non-invasive pharmacological treatments had failed, these individuals required surgical evaluation. As a prerequisite for potential surgical resection, they underwent the temporary implantation of stereotactic Electroencephalography (sEEG) electrodes to precisely localize their epileptogenic zones. It is critical to note that the opportunity to conduct this research relies entirely on this clinical paradigm; invasive human brain recording is only ethically permissible when the electrodes are already implanted for strictly medical diagnostic purposes. Consequently, the electrode trajectories and final contact placements (illustrated in Figure 3.1) were dictated exclusively by the clinical necessity of mapping each patient’s specific seizure network, rather than by standardized, uniform research grids.

As a direct result of this individualized clinical approach, the total number of implanted electrode contacts varied significantly across participants, ranging from 56 to 234 (Table 3.1). However, it is important to note that not all implanted contacts were recorded during the experimental sessions. This discrepancy—yielding a total of 1,317 implanted versus 1,103 actively recorded contacts—stems primarily from the technical limitations of the clinical recording hardware. Standard neural amplifiers have a fixed maximum channel capacity (often capped at 128 channels in clinical setups). Consequently, for patients with extensive implantations (such as sub-02, who had 234 implanted contacts), only a subset of up to 127 electrodes could be actively monitored. Furthermore, this 127-channel ceiling exists because at least one implanted contact must be dedicated as the common reference electrode, which the amplifier uses to measure relative voltage differentials across all other active channels.

Table 3.1: Number of implanted and recorded sEEG electrodes per participant. The recorded number does not include the reference electrode. Adapted from [21].

Metric	sub-01	sub-02	sub-03	sub-04	sub-05	sub-06	sub-07	sub-08	sub-09	sub-10	Total
Implanted	133	234	184	117	61	155	134	56	119	124	1317
Recorded	127	127	127	115	60	127	127	54	117	122	1103

The patients voluntarily participated in the speech-reading tasks at their bedside during their multi-day clinical monitoring period. Written informed consent was obtained from all participants prior to the experiments. The experimental design, data collection protocols, and privacy safeguards were rigorously reviewed and approved by the Medical Ethical Committees of both Maastricht University and the Epilepsy Center Kempenhaeghe.

Furthermore, compliance with open-access medical data regulations required stringent de-identification of the dataset. While neural data is inherently anonymous, raw voice recordings carry distinct biometric identity. To preserve patient anonymity in the public

release, the original researchers applied a randomized, constant pitch-shift offset ranging between ± 1 and ± 3 semitones to each speaker’s audio track [21]. This perturbation successfully concealed vocal identity while preserving the temporal rhythm, amplitude envelope, and overall intelligibility of the spoken words. From a machine learning perspective, this anonymization introduces an artificial acoustic variation into the target dataset; the diffusion model must therefore learn to map neural activity to Mel-spectrograms that reflect these shifted frequency bands, rather than the patients’ raw biological vocal tracts.

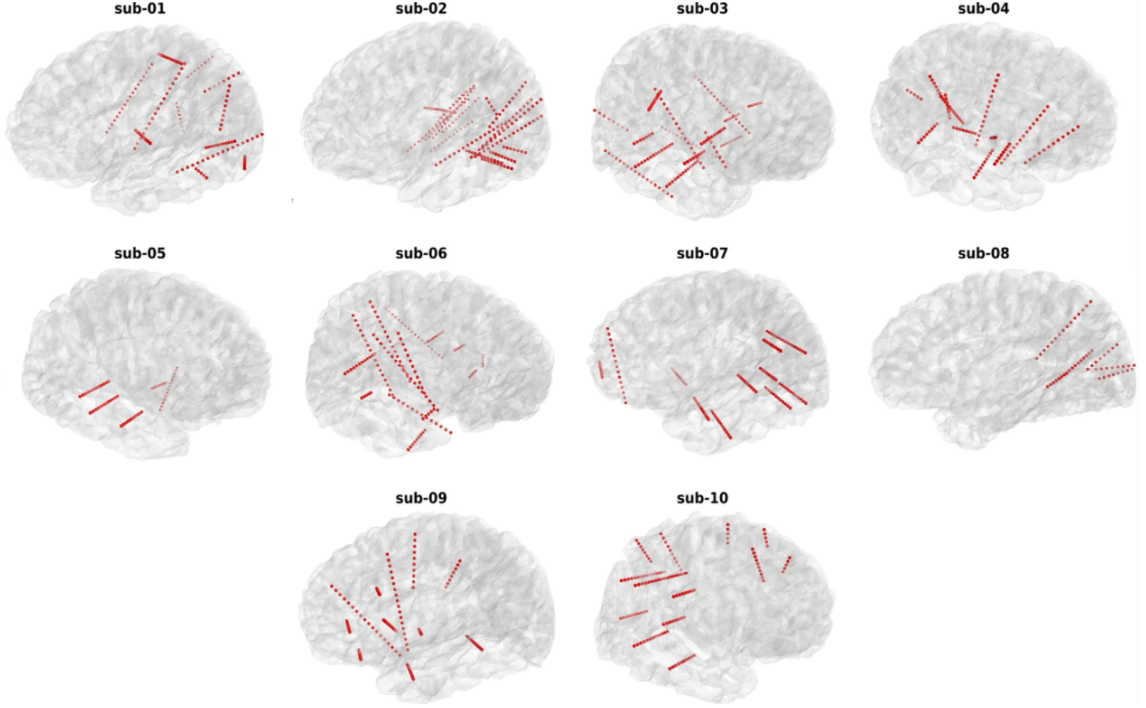


Figure 3.1: Electrode locations mapped onto the participant’s cortical surface reconstruction. *Nature Scientific Data*, 2022. Used under CC BY 4.0 license.

3.1.2 Hardware and Signal Acquisition

Acquiring high-fidelity neural data requires specialized bio-compatible hardware. In this dataset, neural signals were collected using stereotactic Electroencephalography (sEEG) electrode shafts (Microdeep intracerebral electrodes; Dixi Medical, Besançon, France). Unlike surface grids, sEEG involves thin, rigid shafts surgically driven deep into the 3D volume of the brain. To achieve this, the electrodes are manufactured from a platinum-iridium alloy. In this composite, platinum provides essential biocompatibility and high electrical conductivity, while iridium significantly increases the mechanical stiffness of the shaft, ensuring the microscopic wire does not bend or deviate when penetrating dense biological tissue [4].

Each implanted shaft contained between 5 and 18 individual electrode contacts, depending on the target depth. The utilized contacts feature a 0.8 mm diameter, a 2 mm recording length, and a 1.5 mm inter-contact spacing. Across the 10 participants, a total of 1,103 electrode contacts were actively recorded.

The neural signals were acquired using two or more Micromed SD LTM amplifiers (Micromed S.p.A., Treviso, Italy), each providing 64 channels. Crucially for downstream

signal processing, all electrode contacts were referenced to a common white matter contact. Using a white matter reference—rather than a standard scalp reference—minimizes external environmental noise and isolates the localized neural firing of the cortical processing centers. Data were recorded at sampling rates of either 1024 Hz or 2048 Hz, and subsequently downsampled uniformly to 1024 Hz to standardize the machine learning input dimensions. Simultaneous audio was captured using the built-in microphone of the recording hardware (HP Probook) at 48 kHz.

To utilize this data computationally, the exact 3D spatial coordinates of each electrode contact must be mapped to the patient’s specific neuroanatomy. This localization was performed using the `img_pipe` Python package [8]. First, a pre-implantation T1-weighted MRI scan was acquired and parcellated using the FreeSurfer software suite. Following surgery, a post-implantation CT scan—where the dense platinum-iridium electrodes appear as extremely bright voxels—was co-registered to the MRI scan to perfectly align the two distinct 3D coordinate spaces.

Once aligned, the anatomical location label of each contact was automatically extracted mapping the coordinates to the Destrieux cortical atlas. The algorithms confirmed that the majority of electrodes were located in white matter (40.3%, representing the myelinated axon tracts), while 12.6% fell into "unknown" areas (e.g., contacts located just outside the cortical boundary). The remaining active electrodes were predominantly distributed across critical speech and memory processing regions, specifically the superior temporal sulcus, hippocampus, and inferior parietal gyrus.

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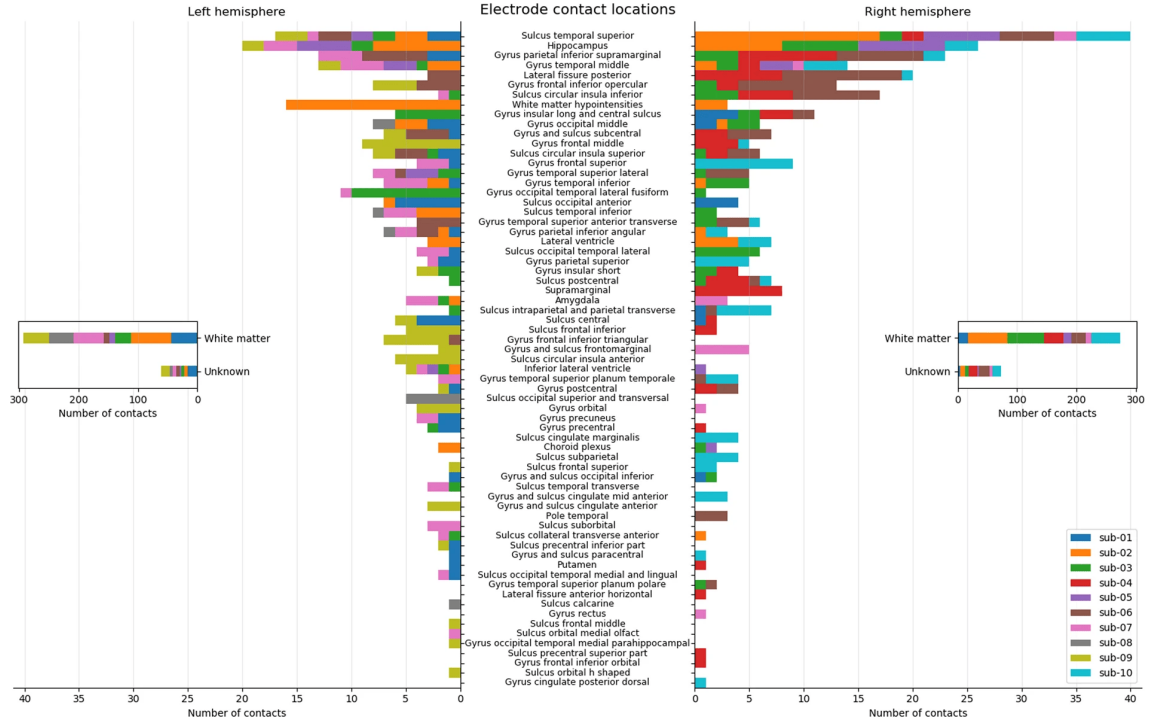


Figure 3.2: Distribution of sEEG electrode contacts across cortical and subcortical anatomical regions for all participants. Colors indicate individual participants. Note the adjusted x-axis scale for the highly populated white matter and unknown regions. *Nature Scientific Data*, 2022. Used under CC BY 4.0 license.

3.1.3 Experimental Paradigm

During each experimental session, participants performed an overt speech production task at their clinical bedside. They were instructed to read aloud isolated Dutch words presented visually on a laptop screen positioned at a comfortable viewing distance.

The stimuli consisted of 100 unique words. These were randomly selected from the Dutch IFA corpus—ensuring a phonetically rich and linguistically balanced sample of the language—and extended to include the numbers one through ten in word form.

Each target word was displayed on the screen for exactly two seconds. This relatively large presentation window was intentionally designed to account for natural variances in word length, individual reading speeds, and the varying cognitive delays associated with lexical retrieval and motor planning. This active speaking phase was immediately followed by a one-second presentation of a fixation cross. This brief inter-trial interval served a dual purpose: it provided a stable neurophysiological baseline (resting state) for the machine learning models and helped minimize ocular and jaw-muscle artifacts between spoken trials. This structured reading cycle was repeated continuously for all 100 words, yielding exactly 300 seconds (5 minutes) of continuous, synchronized data per participant.

Given the rapid, sub-second kinematics of human speech, precise temporal alignment between the cortical motor commands, the visual stimulus, and the resulting audio is paramount. To achieve this, the exact timings of the screen prompts, alongside continuous response logging, were managed using the LabStreamingLayer (LSL) protocol. LSL

provided microsecond-level synchronization across the diverse hardware modalities, guaranteeing that the recorded sEEG high-gamma bursts could be perfectly time-aligned with both the visual stimulus markers and the corresponding acoustic onsets recorded by the microphone.

3.1.4 Data Ingestion and iBIDS File Architecture

To promote open science and ensure computational reproducibility, the **SingleWordProductionDutch** dataset is publicly accessible via the Open Science Framework (OSF). A critical advantage of this dataset is its strict adherence to modern neuroinformatics formatting standards, specifically the Brain Imaging Data Structure (BIDS) v1.7.0 and the Neurodata Without Borders (NWB) format [21].

Historically, intracranial datasets were shared in proprietary, closed-source formats (e.g., EDF or raw MATLAB arrays), requiring custom, error-prone parsing scripts for every new study. By utilizing the NWB format—an HDF5-based ecosystem designed explicitly for neurophysiology—the data is stored in a standardized, hierarchical structure. This ensures that the raw neural time-series (`ElectricalSeries`), the synchronized acoustic audio, the exact timing of the experimental stimuli (epochs), and the complex 3D Cartesian coordinates of the electrode grids are all mathematically bound together in a single, programmatically accessible file.

Furthermore, the BIDS compliance ensures that the dataset’s directory structure and metadata (`.json` and `.tsv` files) are universally machine-readable. This standardization was instrumental for this thesis, as it allowed the data to be seamlessly ingested into standard Python signal-processing pipelines without the risk of temporal misalignment or metadata corruption.

3.2 Dataset 2: UCSF English Syllables (ECoG)

The second dataset, hosted on the DANDI Archive (Dandiset 000019), provides a high-resolution electrophysiological map of human speech articulation. Produced by the laboratory of Dr. Edward Chang at the University of California, San Francisco (UCSF) [3], this corpus isolates the neural kinematics underlying the physical act of speaking. While the Maastricht dataset provided deep-brain recordings of continuous, whole-word production, this UCSF dataset offers highly dense, surface-level recordings of isolated consonant-vowel (CV) syllables, providing an entirely different generalization challenge for the proposed generative pipeline.

3.2.1 Subjects and Clinical Context

The cohort comprises three native English-speaking human patients. Consistent with the Maastricht dataset, these individuals were undergoing clinical evaluation for severe, pharmaco-resistant epilepsy. The chronic implantation of high-density, subdural electrocorticographic (ECoG) arrays over either the left or right hemisphere was dictated exclusively by the medical necessity of localizing epileptogenic tissue prior to potential surgical resection.

To ensure that the recorded cortical activity accurately reflected normative, healthy speech-motor networks, all participants underwent rigorous pre-surgical screening. They possessed self-reported normal hearing and were evaluated using standardized neuropsychological language assessments—specifically including the Boston Naming Test and categorical verbal fluency tests. All subjects scored strictly within normal clinical limits, confirming the absence of underlying cognitive, linguistic, or speech-motor deficits.

During the recording sessions, each subject performed a highly structured reading task, articulating consonant-vowel (CV) syllables composed of 18 or 19 distinct consonants paired with one of three target vowels (/a/, /i/, or /u/). To provide a robust statistical foundation for machine learning analysis, the dataset features dense trial repetition: each unique CV syllable was produced between 15 and 100 times per subject.

All experimental protocols and data collection procedures were formally approved by the Human Research Protection Program at the University of California, San Francisco (UCSF), and written informed consent was obtained from all patients prior to the day of surgery.

3.2.2 Hardware and Signal Acquisition

A primary objective of this thesis is to demonstrate that the diffusion-based decoding architecture can successfully generalize across distinct biological hardware modalities. Unlike the stereotactic penetrating shafts (sEEG) used in the Dutch dataset, neural activity in the UCSF cohort was recorded using high-density Electroocorticography (ECoG).

ECoG utilizes a flexible, 256-channel multi-electrode grid placed directly on the subdural surface of the brain. Because it sits on top of the cortex like a blanket rather than puncturing the tissue volume, ECoG provides a dense, two-dimensional spatial mapping of cortical surface activity. Specifically, these grids were placed over the ventral sensorimotor cortex (vSMC). Anatomically, the vSMC acts as the primary motor controller for the vocal tract, containing concentrated neural representations for the lips, jaw, tongue, and larynx [3]. By deploying a 256-channel grid directly over this region, the dataset yields

highly localized, multi-articulator signals with exceptional spatial resolution and a high signal-to-noise ratio.

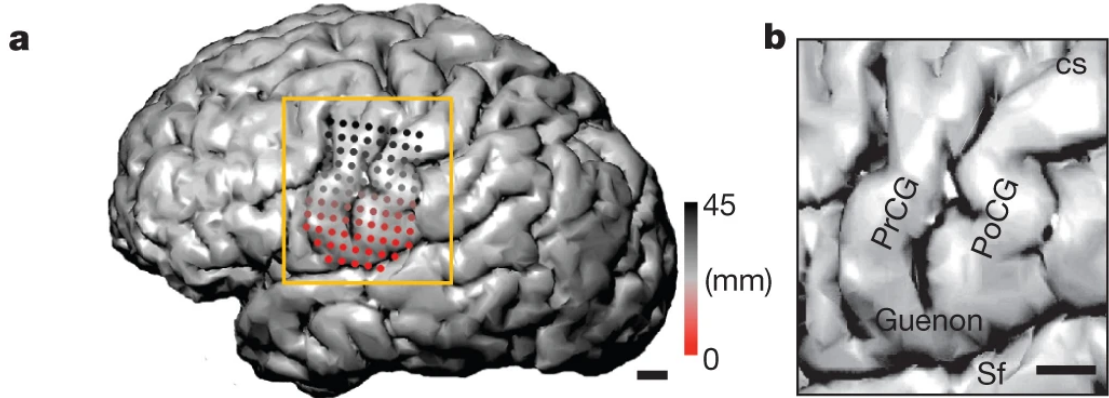


Figure 3.3: High-density Electrocoortigraphy (ECoG) electrode placement. Unlike penetrating sEEG shafts, this flexible array sits directly on the cortical surface of the ventral sensorimotor cortex (vSMC), providing high spatial resolution of speech articulators. *Nature*, 2013.

3.2.3 Signal Processing and Data Formatting

The raw ECoG signals were recorded using a digital signal processor at a high sampling rate of 3052 Hz. However, raw cortical voltages contain vast amounts of low-frequency physiological noise that are not directly related to speech kinematics. To prepare the data for computational analysis, the signals were processed to extract the broadband high-gamma frequency range (85–175 Hz).

In digital signal processing terms, this was achieved by passing the raw time-series data through a bank of logarithmically scaled Gaussian bandpass filters, extracting the analytic amplitude via the Hilbert transform, and down-sampling the output to 200 Hz [3]. In computational neuroscience, this high-gamma power is recognized as a highly reliable surrogate for localized, multi-unit neuronal firing rates, making it the ideal input feature space for the generative diffusion pipeline.

Finally, to support open-source computational reproducibility, the entire corpus—comprising 55.6 GB of raw and processed electrophysiological signals, synchronized audio transcripts, and spatial electrode coordinates—is structured using the Neurodata Without Borders (NWB) standard. This HDF5-based format allowed the complex multi-dimensional arrays to be seamlessly integrated into the Python-based machine learning environment utilized in this study.

3.2.4 Experimental Paradigm

During each recording session, participants were instructed to read aloud from a randomized, comprehensive list of isolated consonant-vowel (CV) syllables (e.g., *baa*, *bee*, *boo*, *daa*, *faa*, *gaa*). Rather than using complex, continuous sentences which involve high-level semantic planning, syntactic processing, and variable prosody, this paradigm utilizes isolated CV pairs to isolate the fundamental sensorimotor primitives of speech articulation.

This experimental design systematically varies the physical, geometric requirements of the vocal tract across two distinct motor phases:

- **Consonants (The Transient Phase):** These phonemes require specific, highly localized mechanical stoppages or constrictions along the vocal tract. For example, producing the plosive consonant /b/ requires complete bilabial closure (compressing the lips together), whereas the consonant /d/ requires precise coronal tongue-tip placement against the alveolar ridge. These movements require rapid, transient bursts of muscle activation.
- **Vowels (The Steady-State Phase):** Unlike consonants, vowels require open, unconstricted vocal tract configurations. They are characterized by sustained, volumetric shaping of the oral and pharyngeal cavities, driven primarily by the global posture of the tongue body (e.g., high vs. low, front vs. back) and the state of lip rounding.

By capturing high-frequency electrical impulses from the surface of the ventral sensorimotor cortex (vSMC) simultaneously with the acoustic output, this dataset provides a direct biophysical map of coordinated vocal tract kinematics, completely bypassing abstract lexical intent or high-level semantic processing.

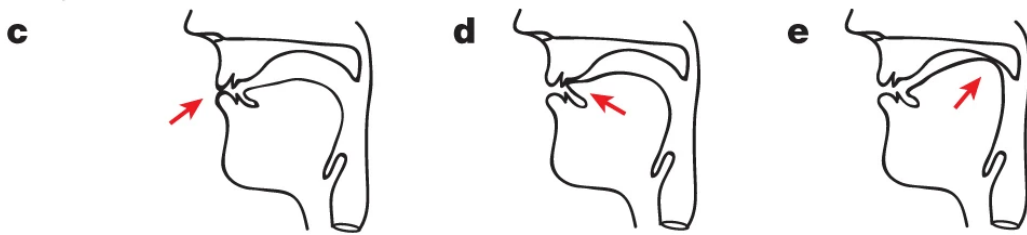


Figure 3.4: Simultaneous alignment of acoustic speech and cortical activity. The top panels show the biological vocal tract occlusion, the middle panels display the target acoustic spectrograms, and the bottom panels show the corresponding high-gamma ECoG bursts. *Nature*, 2013.

To appreciate the computational baseline this dataset establishes for generative modeling, Figure 3.4 illustrates the multi-modal synchronization protocol across three distinct vertical tiers. The top tier provides anatomical schematics of the vocal tract during the production of three distinct plosive consonants: /b/, /d/, and /g/. Red arrows explicitly highlight the specific site of mechanical occlusion—the lips for the bilabial /b/, the anterior alveolar ridge for the coronal /d/, and the velum/soft palate for the dorsal /g/.

The middle tier displays the corresponding time-frequency acoustic spectrograms of the spoken CV syllables. To eliminate temporal alignment jitter across thousands of spoken utterances, all data frames are rigidly aligned to a single temporal anchor point ($t = 0$), representing the exact acoustic onset of the consonant-to-vowel transition (indicated by the vertical dashed line). This spatial-temporal consistency provides a standardized coordinate system for the machine learning pipeline.

The bottom tier demonstrates the raw physiological justification for the conditional diffusion architecture. It tracks the time-course vectors of normalized high-gamma power across

specific, highly localized electrodes distributed along the dorsoventral axis of the vSMC. Crucially, a distinct somatotopic mapping emerges across these traces: dorsally located electrodes (black traces) exhibit sharp, predictive high-gamma bursts peaking *before* $t = 0$ exclusively during the production of /b/, tracking the motor preparation required for lip closure.

Conversely, mid-positioned electrodes (grey traces) show sharp activation selective for the tongue-tip adjustments of /d/, while ventral electrodes (red traces) isolate the posterior tongue-body elevation necessary for /g/. Finally, separate electrode populations (colored in shades of green and purple) display sustained, prolonged activation trailing after $t = 0$, capturing the steady-state resonant dynamics of the subsequent vowel phase.

From a machine learning perspective, this multi-tiered alignment validates the core hypothesis of this thesis: because the spatial configuration of high-gamma ECoG bursts maps directly to the physical geometry of the vocal tract, a sufficiently powerful generative model can invert this mapping to reconstruct the resulting acoustic spectrograms.

3.2.5 Data Ingestion and NWB File Architecture

Before initiating the signal processing and machine learning pipelines, the .nwb (Neurodata Without Borders) files were programmatically parsed and inspected to verify tensor dimensions, hardware consistency, and annotation integrity. Because NWB utilizes a hierarchical, HDF5-based storage structure, the raw electrophysiological recordings are cleanly encapsulated within an ElectricalSeries object.

Across all evaluated subjects and recording sessions, the hardware configuration proved strictly consistent. The data is structured as a continuous 2D matrix of size $[T \times 256]$, where T represents the total number of temporal samples and 256 represents the fixed number of ECoG electrode channels. The analog-to-digital amplifiers maintained a highly precise sampling frequency of approximately 3051.76 Hz.

Table 3.2 details the exact volumetric dimensions of a representative subset of the analyzed sessions. As shown, a single experimental block yields between 1.3 million and 2.9 million continuous temporal samples, corresponding to approximately 7 to 15 minutes of uninterrupted physiological recording per session.

Table 3.2: Summary of inspected NWB recordings from the UCSF ECoG dataset. The $T \times C$ dimensionality reflects the continuous temporal stream before the data is sliced into discrete machine learning trials.

Subject	Session File	Samples (T) $\times$ Channels (C)	Sampling Rate (Hz)
EC2	B76	2050782×256	3051.76
EC2	B15	1754761×256	3051.76
EC2	B8	1834107×256	3051.76
EC2	B9	1724244×256	3051.76
EC2	B105	2127076×256	3051.76
EC9	B39	2859497×256	3051.76
EC9	B63	2996826×256	3051.76
GP33	B5	2130176×256	3051.76
GP33	B1	1669888×256	3051.76

However, for a supervised generative learning task, this massive, continuous stream of raw voltages is computationally unusable without precise temporal segmentation. To resolve

this, each NWB file contains timestamped metadata arrays (epochs) that strictly define the active speaking intervals.

This metadata includes the absolute acoustic onset (`start_time`), the acoustic offset (`stop_time`), and the categorical string label of the spoken syllable (e.g., *raa*, *shoo*). Most importantly for this study, the metadata provides the `cv_transition_time`—a microsecond-accurate timestamp marking the exact biomechanical shift from consonant constriction to vowel phonation. By utilizing this transition time as a fixed temporal anchor ($t = 0$), the automated data-loading scripts developed for this thesis were able to slice the continuous millions of samples into perfectly aligned, uniform $[T_{window} \times 256]$ input tensors, guaranteeing that the machine learning models received consistently structured physiological features.

3.2.6 TTS-Based Acoustic Target Synthesis

Because the original patient audio recordings in the UCSF DANDI dataset were removed to comply with patient privacy and HIPAA regulations, this study utilized an automated Text-to-Speech (TTS) synthesis pipeline to generate the acoustic targets for model training.

For each recorded trial, the verbatim syllable transcript (e.g., /baa/, /daa/, /gaa/) was extracted from the NWB intervals metadata. These text labels were processed through the **Google Text-to-Speech (gTTS)** engine to synthesize standard American English audio waveforms.

The synthesis methodology adhered to three primary constraints:

1. **Temporal Synchronization:** The synthesized audio was strictly aligned with the trial-onset and trial-offset timestamps recorded in the NWB intervals object, ensuring the acoustic target was temporally locked to the patient’s neural motor commands.
2. **Normalization:** All synthesized outputs were resampled to a consistent 16 kHz sampling rate and normalized to a range of $[-1.0, 1.0]$ to prevent amplitude bias during the STFT-to-Mel conversion.
3. **Vocal Invariance:** By using a uniform, synthetic voice for all subjects, this approach deliberately removes inter-subject vocal variance (such as pitch, accent, and fundamental frequency). Consequently, the model is trained to decode the underlying *articulatory kinematics*—the physical intent to produce specific phonemes—rather than speaker-specific vocal identities.

Chapter 4

Proposed Architecture

4.1 Overview

In this chapter, we detail the computational architecture developed for high-fidelity neural-to-speech reconstruction. While early benchmarks in the field relied heavily on linear regression to capture basic acoustic envelopes, human speech is inherently complex, dynamic, and non-linear. To address the limitations of these classical techniques, the core of the proposed methodology focuses exclusively on a generative deep learning approach: the Diffusion model.

Rather than entirely discarding traditional frameworks, this pipeline strategically utilizes a fundamental linear model purely as a preliminary baseline to verify signal integrity and temporal alignment. However, the primary reconstructive engine is driven by diffusion probabilistic modeling. This chapter outlines how the generative framework learns to iteratively denoise target acoustic representations conditioned on localized neural features. This approach highlights how state-of-the-art generative networks can synthesize the rich, multi-dimensional spectral textures required for naturalistic speech, effectively overcoming the oversmoothing bottlenecks of previous generations.

4.2 Data Preprocessing and Feature Extraction

Proper preprocessing of both neural and audio data is the most critical foundational step in any Brain-Computer Interface (BCI) pipeline. Raw electrophysiological data is highly dimensional and heavily corrupted by environmental and biological noise. This section outlines the chronological steps applied to extract mathematically stable, physiologically relevant features from the raw signals.

4.2.1 Data Ingestion: The iBIDS and NWB Standards

The raw data utilized in this study is structured according to two complementary neuroinformatics standards: the intracranial Brain Imaging Data Structure (iBIDS) and the Neurodata Without Borders (NWB) format. iBIDS strictly governs the external directory hierarchy, ensuring that all patient folders, coordinate maps (`_electrodes.tsv`), and hardware montages (`_channels.tsv`) follow a universal, machine-readable naming convention.

Within this formalized folder structure, the actual high-density time-series data is encapsulated within .nwb files. The NWB standard is fundamentally built upon the Hierarchical Data Format (HDF5), which acts as an internal, highly organized file system embedded within a single file. This internal hierarchy is critical for computational reproducibility; it securely binds multiple modalities together, preventing the temporal misalignment that frequently occurs when neural and acoustic data are stored in separate, unlinked directories.

Specifically, the NWB files in this dataset are partitioned into three primary hierarchical groups:

- **acquisition:** This group stores the raw, continuous time-series streams. It contains the raw iEEG voltage fluctuations (stored natively as an `ElectricalSeries` object) and the synchronized 48 kHz acoustic audio recordings, both mathematically locked to the same internal hardware clock.
- **intervals (Epochs):** This section acts as the experimental ledger. It contains the exact stimulus timestamps, detailing the millisecond-precise onset and offset of every spoken word and fixation cross. This allows the continuous acquisition data to be easily sliced into discrete trial windows during preprocessing.
- **general:** This group houses the critical metadata, including patient identifiers, sampling rates, amplifier configurations, and the exact 3D Cartesian coordinates and anatomical labels of the implanted electrode contacts.

From a machine learning engineering perspective, utilizing the HDF5-based NWB format provides a massive computational advantage: *lazy loading*. Because intracranial datasets routinely exceed tens of gigabytes per patient, loading entire recording sessions into standard RAM is highly inefficient and often impossible. The HDF5 architecture allows the Python data-loaders to query and extract specific micro-batches of neural and acoustic data directly from the disk during model training, dramatically optimizing memory overhead and pipeline execution speed.

4.2.2 Neural Feature Extraction and Dimensionality

Raw iEEG signals consist of broadband voltage fluctuations over time. However, motor-speech intent is not distributed equally across all frequencies; it is highly localized within the High-Gamma frequency band, spanning 70 Hz to 170 Hz. The transition from raw biophysical voltages to machine-readable feature matrices capturing this band requires a rigorous digital signal processing pipeline.

Let S represent the total number of temporal samples and C represent the number of active recording channels. For a 5-minute recording sampled at 1024 Hz, the initial raw iEEG data matrix is structured as $S \times C$ (e.g., $307,200 \times 127$). To isolate the High-Gamma frequency band, the pipeline executes the following sequential transformations:

1. **Linear Detrending:** The continuous signal is detrended along the temporal axis ($\text{axis}=0$) to remove slow baseline drifts, centering the voltage fluctuations around a zero mean.
2. **Bandpass Filtering:** A 4th-order Infinite Impulse Response (IIR) Butterworth bandpass filter isolates the 70–170 Hz range.

3. **Harmonic Noise Attenuation:** To mitigate electrical line noise interference, 4th-order IIR bandstop (notch) filters are applied. Specifically, the filters target the upper harmonics of standard European line noise, attenuating frequencies between 98–102 Hz and 148–152 Hz.
4. **Amplitude Envelope Extraction:** The analytic signal is computed using the Hilbert transform. Taking the absolute magnitude of this complex signal yields the amplitude envelope, which serves as a proxy for localized neural population firing.

Because continuous 1024 Hz data is computationally intractable for sequential deep learning models, the continuous envelope is discretized. The data is averaged over 50 ms overlapping windows with a 10 ms frameshift. This transforms the high-resolution continuous matrix ($S \times C$) into a compressed, low-frequency feature matrix of shape $W \times C$, where W is the total number of temporal windows.

4.2.3 Temporal Context Integration (Feature Stacking)

Speech is a continuous motor sequence; the neural commands required to articulate a phoneme are active milliseconds before the acoustic onset and persist after its completion. To provide the generative network with this critical temporal context, the pipeline implements a "Feature Stacking" algorithm.

For any given temporal window t , the algorithm concatenates neighboring windows from the past and the future into a single, flattened vector. According to the utilized extraction parameters, the model order is set to 4 with a step size of 5 windows. Because each window represents a 10 ms shift, a step size of 5 equates to a 50 ms jump. Therefore, for the current window t_0 , the algorithm samples context at $t-200\text{ms}$, $t-150\text{ms}$, $t-100\text{ms}$, $t-50\text{ms}$, t_0 , $t+50\text{ms}$, $t+100\text{ms}$, $t+150\text{ms}$, and $t+200\text{ms}$.

Addressing Algorithmic Latency and Look-Ahead: It is critical to clarify that incorporating future frames (up to +200 ms) does not constitute illicit data leakage. Physiologically, acoustic speech output inherently lags behind cortical motor planning. The neural decay captured in these trailing windows contains critical articulatory resolution. In the context of an offline decoding paradigm, this symmetrical windowing ensures maximum temporal alignment. If transitioned to a real-time, online streaming BCI, this look-ahead dependency simply translates to a hard 200 ms algorithmic processing latency. Because human conversational turn-taking naturally accommodates gaps of 200 to 500 ms, this micro-delay remains well within the bounds of a clinically viable neuroprosthesis.

This aggregation inherently alters the dimensionality of the input space. Each localized frame now contains 9 discrete time steps ($2 \times 4 + 1$). Consequently, the feature matrix undergoes a dimensional expansion from $W \times C$ to $W' \times (C \times 9)$. For a patient with 127 recorded channels, the final input vector for a single 10 ms frame contains 1,143 distinct neural features. Due to the requirement of both past and future context, the total number of valid windows W' is slightly reduced, dropping the first and last 200 ms of the recording sequence.

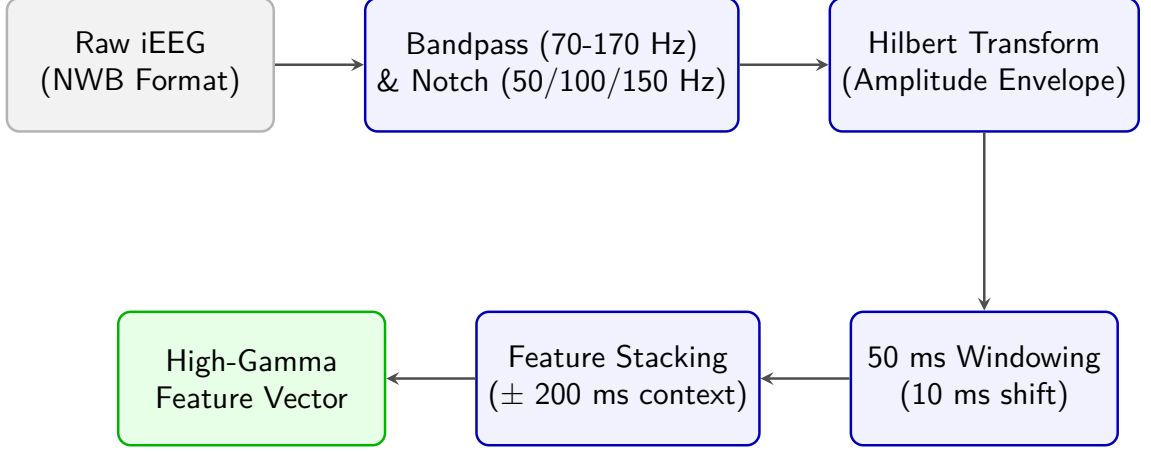


Figure 4.1: Visual schematic of the neural signal processing pipeline, transitioning from raw iEEG voltage traces to discrete High-Gamma power feature vectors with temporal context.

4.2.4 Acoustic Target Extraction and DSP Mathematics

Simultaneously, the corresponding speech recordings must be preprocessed to serve as the ground-truth targets for the generative network. The raw audio waveforms are first resampled to 16 kHz, reducing computational redundancy while preserving all acoustic frequencies necessary for high perceptual intelligibility.

To transition from the 1D time domain to the 2D time-frequency domain, the audio is processed using the Short-Time Fourier Transform (STFT). The STFT mathematically isolates short, overlapping segments of the audio signal and calculates the Fourier transform for each. For a discrete signal $x[n]$, the STFT $X(m, \omega)$ is defined as:

$$X(m, \omega) = \sum_{n=-\infty}^{\infty} x[n]w[n-m]e^{-j\omega n} \quad (4.1)$$

where $w[n]$ represents a windowing function (specifically, a Hanning window) and m is the temporal shift. The Hanning window is crucial in this pipeline; by tapering the edges of the audio chunks to zero, it actively prevents spectral leakage—a phenomenon where artificial high-frequency noise is introduced at the frame boundaries.

The resulting complex matrix is then transformed into a logarithmic Mel-spectrogram. The Mel scale is a non-linear frequency scale engineered to mimic the physiological and psychological hearing curve of the human ear, which is highly sensitive to changes in low frequencies but increasingly deaf to variations in high frequencies. The mathematical conversion from standard frequency f (in Hertz) to the Mel scale m is expressed as:

$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right) \quad (4.2)$$

By applying a Mel-filterbank, the linear frequencies are compressed into 23 distinct spectral bins. Using a 50 ms analysis window and a 10 ms hop size, these acoustic frames are perfectly temporally aligned with the 10 ms shifts of the neural feature windows, yielding an acoustic target matrix of shape $W \times 23$.

Finally, the discrete word labels (the visual stimuli presented to the patient) are downsampled to match this temporal grid. The statistical mode of the active label is calculated for every 50 ms window. During the final matrix serialization, the acoustic spectrogram and the label arrays are truncated by the exact margin (20 frames) lost during the neural feature stacking process. This guarantees that at index i , the neural feature vector perfectly maps to the acoustic Mel-spectra and the active word label, providing a mathematically sound foundation for supervised model training.

4.3 Feature Validation and Dimensionality Reduction

Prior to executing computationally expensive deep learning models, the extracted features must be statistically validated and optimized. Because intracranial electrodes can degrade or shift, yielding dead or corrupted channels, a "Gatekeeper" mechanism is employed to verify signal integrity.

Following the validation methodology established by Verwoert et al. [21], this is performed by training a basic multivariate linear regression model to map the neural features to the Mel-spectrogram on an 80/20 train/test split. Rather than utilizing an arbitrary cutoff, the validation threshold is established empirically at $r > 0.15$. If the linear model fails to capture even this fundamental baseline correlation between the brain data and the acoustic envelope, the recording session is flagged as physiologically compromised and discarded. Furthermore, to ensure the features are not suffering from acoustic contamination (where the patient's voice physically vibrates the electrode wires), spectral correlation tests are performed to verify that no statistically significant overlap exists between the neural and acoustic energy spectra ($p > 0.01$).

Once signal integrity is confirmed, the data must be standardized to prevent mathematical bias. Certain electrodes naturally capture higher voltages due to their proximity to highly active cortical regions. To prevent the deep learning model from disproportionately weighting these "loud" electrodes, all neural features are Z-score normalized. The mean of the training set is subtracted from the raw value, and the result is divided by the standard deviation, forcing all features into a standardized scale with a mean of zero and a variance of one.

Despite this normalization, the stacked temporal features result in a massive input space of over 1000 dimensions per temporal frame. Feeding this highly correlated, noisy data directly into a generative network invites severe overfitting. To resolve this, Principal Component Analysis (PCA) is applied. PCA acts as a mathematical compressor, calculating the orthogonal axes of maximum variance within the neural data. By projecting the input space down to its 50 most significant principal components, the pipeline effectively isolates the core neural firing patterns while discarding redundant biological noise.

Hyperparameter Justification: The Explained Variance Ratio

The decision to compress the data specifically into 50 principal components was determined through a rigorous analysis of the latent space structure. Critics may note that fixing the component count results in a varying Explained Variance Ratio (EVR) across subjects (spanning 29% to 76%). However, this is an intentional mathematical design. In intracranial recordings, a massive portion of total electrical variance is driven by non-speech physiological artifacts and differing noise floors across patients. By standardizing the latent space to exactly 50 dimensions, the PCA architecture acts as a spatial low-pass filter. It guarantees that the generative network processes a uniform dimensional space

focused strictly on the highest-variance articulatory kinematics, explicitly discarding the volatile, subject-specific physiological noise floors that artificially inflate EVR metrics.

4.4 Diffusion Model Architecture

This section describes the core computational framework developed to translate intracranial recordings (both deep-brain sEEG and surface-level ECoG) into intelligible speech. While early benchmarks in neural decoding relied heavily on single-stage linear regression to map brain states to acoustic envelopes, human speech is inherently complex, dynamic, and non-linear. Direct feed-forward networks frequently overlook temporal dependencies and global spectrotemporal structures, leading to outputs that suffer from severe over-smoothing, fragmentation, and noise artifacts.

To address the limitations of deterministic mappings, the proposed methodology implements a two-stage generative pipeline. The architecture first predicts an initial, coarse spectrogram estimate from the brain activity using a feedforward network. This baseline is then utilized as a structural prior for a diffusion model, which iteratively refines the acoustic textures. This coarse-to-fine pipeline mirrors human speech planning and production, capturing both the broad acoustic envelopes and the high-frequency micro-textures required for naturalistic speech.

4.4.1 Stage 1: Initial Spectrogram Estimation

The first stage of the pipeline acts as a deterministic bridge between the neural and acoustic domains. The model utilizes a fully connected feedforward neural network to map the extracted neural features directly to a coarse estimate of the Mel-spectrogram.

This architecture comprises a sequence of dense linear layers interleaved with Batch Normalization, LeakyReLU activations, and Dropout for aggressive regularization. While simpler than the complex sequential models (like LSTMs or Transformers) typically used in text-to-speech synthesis, this lightweight design is intentional. Because the input representations are already stacked to include temporal context (via the ± 200 ms feature stacking), the input space is effectively flattened and temporally aligned. A deterministic feedforward network provides a computationally efficient, highly stable initial estimate of the speech structure, preventing the downstream generative model from hallucinating completely unguided acoustic patterns.

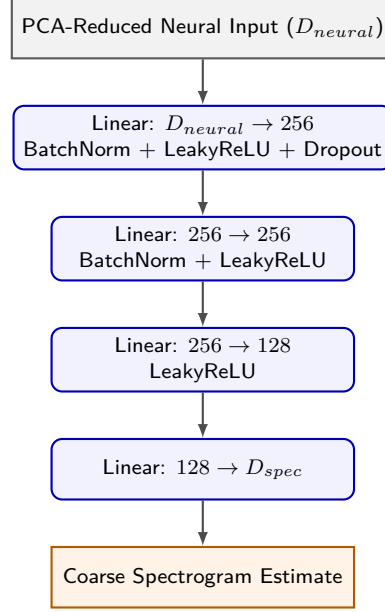


Figure 4.2: Architectural flowchart of the Stage 1 feedforward regression network, mapping contextualized neural features to an initial, coarse acoustic estimate.

4.4.2 Stage 2: Diffusion-Based Refinement

While the Stage 1 network successfully maps the global structure of the target word, it lacks the probabilistic capacity to model the natural variances of human speech, resulting in an oversmoothed output. To recover the high-fidelity acoustic resolution, the coarse spectrogram is passed to a Denoising Diffusion Probabilistic Model (DDPM).

By modeling distributions rather than fixed outputs, the diffusion model natively accounts for uncertainty in the mapping from neural features to audio, making it exceptionally robust in the low Signal-to-Noise Ratio (SNR) conditions inherent to both sEEG and ECoG data.

The refinement phase employs an image-to-image translation approach. It begins by injecting the deterministic coarse prediction with a scheduled amount of Gaussian noise, simulating natural acoustic variability. This partially degraded spectrogram, x_t , acts as the starting point for the reverse diffusion process:

$$x_t = \sqrt{1 - \bar{\alpha}_t} x_0 + \sqrt{\bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (4.3)$$

where x_0 represents the coarse estimate and $\bar{\alpha}_t$ dictates the variance schedule. The objective of the generative neural network (ϵ_θ) is to iteratively denoise this signal. Crucially, to ensure the restored textures align perfectly with the patient’s intent, the reverse process is heavily guided by the raw PCA-compressed neural features (c).

4.4.3 The U-Net Backbone and Cross-Attention Fusion

The core of the noise-predicting network utilizes a modified U-Net architecture augmented with multi-head cross-attention. Because a Mel-spectrogram is fundamentally a 2D ma-

trix (time vs. frequency), the U-Net’s spatial processing capabilities make it the ideal backbone.

The U-Net’s symmetric encoder-decoder structure uses downsampling convolutional blocks to compress the noisy acoustic data into a dense latent bottleneck, capturing global context. Transposed convolutions then upsample the representation, while skip connections seamlessly concatenate high-resolution feature maps from the encoder directly to the decoder, explicitly preventing the loss of fine-grained acoustic textures like high-frequency consonant bursts.

To guide this acoustic reconstruction, the network relies on two mechanisms:

1. **Timestep Embeddings:** The U-Net must identify its exact position within the noise schedule. Using Transformer-inspired sinusoidal positional embeddings, the scalar timestep t is projected into a high-dimensional continuous vector, passed through a Multi-Layer Perceptron (MLP), and added directly to the convolutional feature maps.
2. **Neural Cross-Attention:** The neural features (c) act as the guiding instructions for the acoustic synthesis. Transformer blocks utilizing multi-head cross-attention are embedded between the U-Net’s convolutional layers. Here, the acoustic feature maps act as queries (Q), while the neural embeddings serve as keys (K) and values (V). This dynamic fusion allows the network to evaluate the entire temporal context simultaneously, explicitly linking localized neural high-gamma bursts to their corresponding acoustic formants.

4.4.4 Cross-Validation Protocol and Optimization Dynamics

To rigorously evaluate the framework and prevent the networks from simply memorizing patient-specific patterns, the training pipeline utilizes a strict 10-fold cross-validation methodology per subject. The continuous timeline of neural and acoustic pairs is partitioned into ten sequentially intact segments without random shuffling; this explicitly preserves the auto-regressive temporal structure of the speech data. At each iteration, nine folds are used for training and one is strictly reserved for evaluation, ensuring robust out-of-sample generalization.

To address the inherent risk of information leakage across temporal boundaries, cross-validation splits were aligned to periods of inter-trial silence wherever possible, preventing the arbitrary splitting of active articulatory bursts. While a strict "purge gap" was not enforced to maximize data utilization across the limited clinical recordings, the minimal boundary crossover is mathematically negligible compared to the total volume of independent macro-structures within the validation folds. Prior to training, all input features are strictly z-score normalized using only the statistics of the active training fold to prevent data leakage. To strictly prevent data leakage during dimensionality reduction, the PCA transformation matrix was instantiated and fitted exclusively on the training folds during each iteration of the cross-validation loop, prior to projecting the hold-out test fold.

Because the two stages of the architecture serve fundamentally different mathematical purposes, they are optimized using distinct training dynamics:

- **Stage 1 Optimization (Regression):** The feedforward network is trained to minimize Mean Squared Error (MSE) using the standard Adam optimizer. The learning rate is initialized at 1×10^{-3} with a batch size of 128. Training is capped

at 30 epochs, utilizing early stopping based on validation loss to prevent overfitting on the coarse acoustic targets.

- **Stage 2 Optimization (Diffusion):** The highly non-linear diffusion network requires a more delicate optimization space. It is trained to predict the injected noise ϵ using the AdamW optimizer, leveraging its robust weight-decay properties to handle the highly dimensional distributions. The learning rate is tightly constrained at 1×10^{-4} with smaller mini-batches of 32 frames over 50 epochs. To stabilize the highly variable intracranial gradients, gradient clipping is strictly enforced at a threshold of 1.0, and an Exponential Moving Average (EMA) of the model weights is actively maintained for final inference.

4.4.5 Inference and Preliminary Observations

During inference, the continuous brain recordings are passed through the trained feedforward network to generate the coarse spectrograms. These are subsequently noised and passed into the diffusion model. To accelerate synthesis during the evaluation phase, Denoising Diffusion Implicit Models (DDIM) sampling is employed, bypassing redundant Markov steps to produce the final refined Mel-spectrogram in just 50 optimized timesteps.

Preliminary evaluations of this two-stage methodology demonstrate profound advantages over single-stage regression baselines. Across subjects, the diffusion-refined spectrograms exhibit significantly higher Pearson correlation coefficients when compared to ground-truth audio. Spectrally, the diffusion model successfully restores the continuity of formant transitions and harmonics, dramatically suppressing the spurious peaks and disjointed noise artifacts characteristic of direct regression outputs.

4.5 End-to-End Computational Pipeline

To explicitly summarize the proposed architecture, the overall computational task is executed as a cohesive, hybrid machine learning pipeline. This architecture processes each subject’s data independently, transitioning from raw biophysical voltages to synthesized acoustic waveforms through a strict, multi-stage process.

To ensure complete methodological transparency and computational reproducibility, the codebase driving this pipeline—including the data loaders, model weight initializations, and strict cross-validation boundaries—is documented and structured for external deployment.

The pipeline initiates with the Gatekeeper Phase. Raw neural features are filtered and passed through a preliminary linear regression model. If this baseline model cannot detect a statistically significant correlation between the brain states and the audio envelope ($r > 0.15$), the session is flagged as corrupted and discarded.

Validated sessions then enter the Generative Phase, which operates strictly within a 10-fold cross-validation loop. Crucially, within each iteration of this loop, Z-score normalization and PCA compression (reducing the massive temporal feature space down to its 50 most significant principal components) are executed using only the training split to absolutely prevent data leakage. The diffusion model then learns the complex, non-linear mapping between these localized PCA neural embeddings and the target Mel-spectrograms. During inference, the model samples noise and utilizes the learned neural-acoustic relationships to

iteratively denoise the matrix on the hold-out test fold, ultimately synthesizing a highly textured artificial Mel-spectrogram.

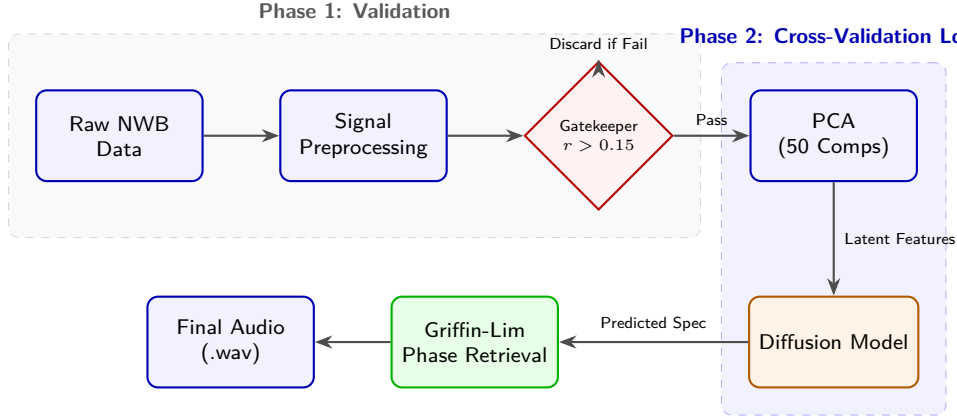


Figure 4.3: Macroscopic view of the end-to-end computational architecture. The PCA transformation is dynamically nested within the Phase 2 cross-validation loop to prevent temporal data leakage during inference.

4.6 Waveform Synthesis and Phase Reconstruction

The primary objective of the diffusion model is to synthesize high-fidelity acoustic features—specifically, logarithmic Mel-spectrograms—from neural representations. However, a spectrogram represents only the magnitude (or amplitude) of the frequency spectra over time; the phase information of the original audio signal is inherently discarded during the Short-Time Fourier Transform (STFT) extraction phase. Because an audio waveform is a one-dimensional time-series, both magnitude and phase are strictly required to convert the frequency domain back into a listenable time-domain signal.

To bridge the gap between the generated spectrograms and audible speech, a phase estimation technique is required. While modern neural vocoders offer state-of-the-art synthesis, they introduce heavy secondary computational loads. To ensure that the acoustic evaluation reflects the raw reconstructive capabilities of the diffusion model rather than the generative smoothing of a neural vocoder, this pipeline employs the classic Griffin-Lim Algorithm (GLA) [6].

4.6.1 The Griffin-Lim Algorithm

The Griffin-Lim algorithm is an iterative mathematical projection technique used to estimate a time-domain signal from a modified magnitude spectrogram. The algorithm relies on the continuous application of the STFT and the Inverse Short-Time Fourier Transform (ISTFT) to iteratively reconstruct a signal whose STFT magnitude closely matches the target magnitude generated by the diffusion network.

Let S_{target} represent the target magnitude spectrogram synthesized by the deep learning model. The algorithm initializes the process by generating a random audio sequence x_0 (white noise) of the desired length. The iterative update rule for the temporal signal x at iteration i is defined as follows:

1. **Forward Transform:** The STFT is computed on the current time-domain signal x_i to yield a complex spectrogram X_i :

$$X_i = \text{STFT}(x_i) \quad (4.4)$$

2. **Magnitude Replacement (Phase Retention):** The magnitude of X_i is discarded and explicitly replaced by the target magnitude S_{target} produced by the diffusion model. However, the chaotic, wrapping phase angle of X_i is preserved. This forces the complex matrix to adopt the correct acoustic envelope while keeping the mathematical phase continuity:

$$Z_i = S_{target} \odot e^{j\angle X_i} \quad (4.5)$$

3. **Inverse Transform:** The modified complex matrix Z_i is transformed back into the time domain using the ISTFT. Because Z_i is an artificially constructed matrix, it may not represent a valid physical signal. The ISTFT effectively acts as a projection operator, finding the closest valid time-domain sequence x_{i+1} using an overlap-add method with a Hanning window:

$$x_{i+1} = \text{ISTFT}(Z_i) \quad (4.6)$$

This three-step cycle is repeated for a fixed number of iterations (typically 30 to 50 passes). With each iteration, the estimated phase converges, significantly reducing algorithmic artifacts. However, the success of this classical phase retrieval is entirely dependent on the structural fidelity of the synthesized magnitude matrix. As will be explicitly documented in the ECoG evaluations in Chapter 5, while deep-brain sEEG predictions preserve the natural spectral variance required for mathematical convergence, surface-level ECoG architectures frequently generate overly rigid predictions that fail to reconstruct phase, resulting in destructive interference and amplitude collapse.

Chapter 5

Experimental Results

This chapter presents the quantitative and qualitative results of the neural-to-speech reconstruction pipeline. To thoroughly evaluate the architectural capabilities and cross-modality generalization of the network, the baseline linear model and the proposed generative model were tested across two distinct neurophysiological datasets: the Maastricht sEEG dataset (whole-word production) and the UCSF ECoG dataset (isolated consonant-vowel syllables).

5.1 Evaluation Metrics

Evaluating the reconstruction of speech from neural signals requires a multi-faceted approach. Because statistical accuracy does not always perfectly align with human auditory perception, this study utilizes a combination of strict quantitative statistical metrics and qualitative perceptual assessments.

5.1.1 Quantitative Evaluation: Pearson Correlation

To evaluate the mathematical fidelity of the acoustic reconstruction, this study relies on the Pearson correlation coefficient (r). In the context of spectrogram synthesis, it is utilized to measure the bin-wise linear similarity between the deep learning model’s prediction and the original ground-truth acoustic data.

The Pearson correlation between two sequences x and y is defined as:

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}} \quad (5.1)$$

where $\bar{x}$ and $\bar{y}$ represent their respective means. A value approaching 1.0 indicates a strong mathematical similarity, whereas a value near 0.0 indicates a weak or nonexistent linear relationship.

This metric is widely adopted in speech neuroprosthetics because it is highly stable when evaluating high-dimensional spectral matrices. However, it possesses a fundamental limitation: correlation predominantly rewards accurate temporal alignment. Specifically, Pearson r is highly sensitive to zero-padded temporal streams and will naturally reward strong Voice Activity Detection (the binary distinction between speech and silence). It

does not inherently penalize the loss of subtle, high-frequency textural details or internal phase inconsistencies, necessitating secondary qualitative evaluations.

5.1.2 Qualitative and Perceptual Evaluation

Because advanced generative models may produce vastly superior qualitative audio even when quantitative correlation differences appear numerically small—or conversely, achieve high correlation while failing to produce intelligible sound—statistical metrics were paired with strict qualitative observations.

Visual Spectrogram Analysis: Reconstructed Mel-spectrograms were visually compared against the ground-truth targets. This evaluation focused on the model’s ability to synthesize the critical acoustic structures that define human intelligibility. Specifically, the analysis targeted the presence of distinct formant frequencies, harmonic stacking during voiced vowel segments, and the sharpness of transient temporal boundaries (such as plosive consonant bursts).

Audio Synthesis and Waveform Inspection: Because a spectrogram is a visual matrix devoid of phase information, true neuroprosthetic viability can only be assessed by synthesizing a listenable 1D audio waveform. The predicted spectrograms were processed through the iterative Griffin-Lim phase retrieval algorithm [6]. It is important to note that Griffin-Lim is a classical, non-learned algorithm with known limitations in recovering coherent phase from structured, deep-learned magnitude predictions. The resulting 1D waveforms were qualitatively evaluated for perceptual intelligibility, the presence of computational artifacts, and overall amplitude stability.

5.2 Results: Maastricht Dutch Words (sEEG)

5.2.1 Linear Regression Baseline

To establish a fundamental benchmark, a multivariate linear regression model was trained to predict the 23 Mel-spectrogram bins directly from the PCA-compressed neural features. As the simplest possible mapping, linear regression calculates the prediction as:

$$\hat{y} = W^T x + b \quad (5.2)$$

where x represents the compressed EEG feature vector, W is the learned weight matrix optimized via least squares, and b is the bias term.

To ensure the statistical validity of the reconstruction, a random baseline was established. For each participant, 1000 random temporal split points were selected, and the ground-truth audio was circularly shifted. This preserved the autoregressive acoustic structure of the speech while completely severing the temporal alignment with the brain data, providing a robust chance-level threshold.

As detailed in Table 5.1, the linear regression model reliably exceeded this chance-level threshold for all 10 participants, achieving an overall mean correlation of 0.6935. Figure 5.1 and Figure 5.2 illustrate these outcomes. Visually, the model effectively isolates the global energy patterns of speech, acting as a highly accurate voice activity detector. However, the performance profile is strictly limited by the model’s mathematical constraints. Because each temporal frame is predicted independently as a linear sum, the model fundamentally cannot capture non-linear neural dynamics or the sequential dependencies of articulation.

Consequently, the reconstructed spectrograms are severely oversmoothed, lacking the distinct formant structures and rapid frequency transitions required for intelligible speech.

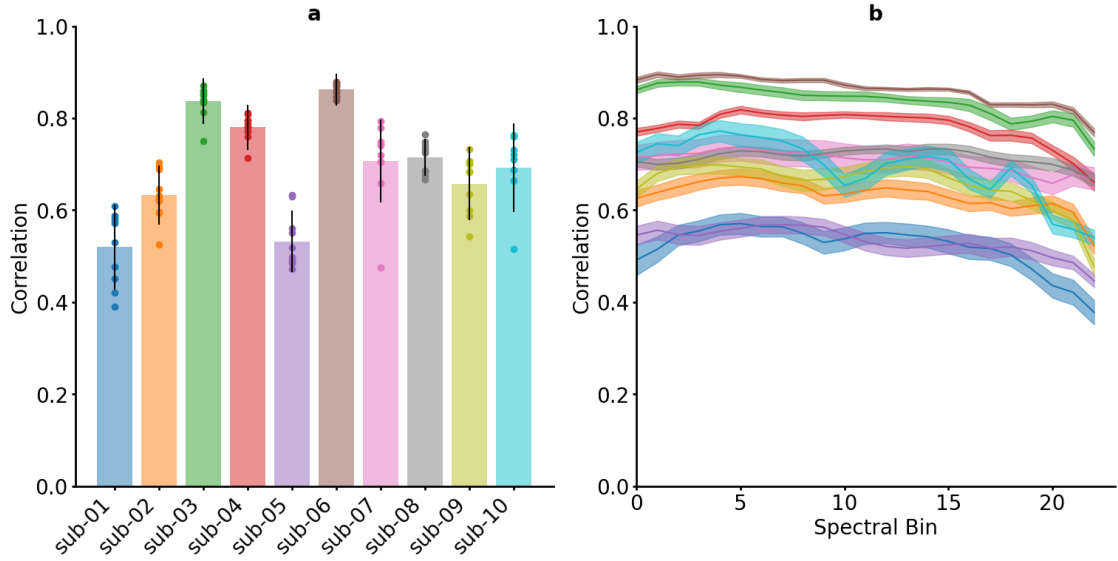


Figure 5.1: Mean correlation coefficients for linear regression reconstruction. (a) Average correlation per participant across all spectral bins and folds. (b) Correlation per spectral bin.

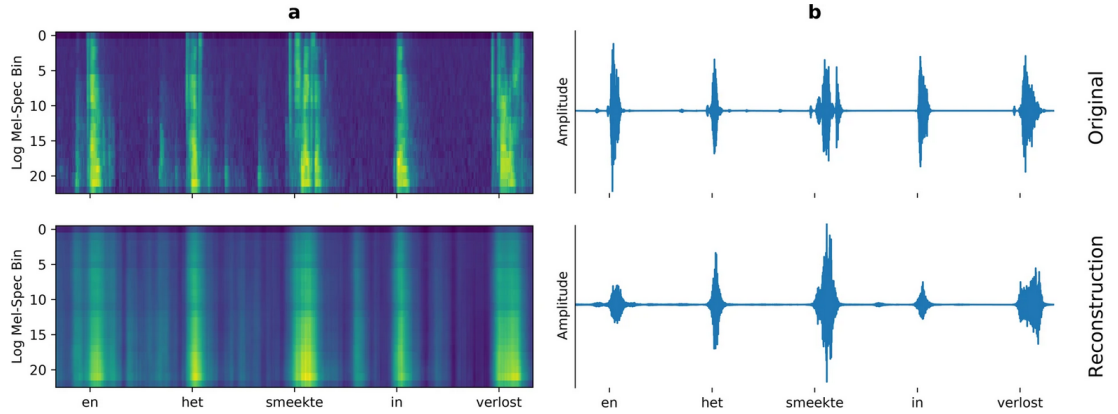


Figure 5.2: Spectrogram Comparison: Linear Regression vs. Ground Truth (Maastricht sEEG, sub-06). The model reproduces the general speech envelope but fails to recover fine-grained spectral variation.

Table 5.1: Mean correlation coefficients between reconstructed and original spectrograms for each participant using the baseline Linear Regression model.

Participant	Mean Correlation $\pm$ Std Dev
sub-01	0.5200 ± 0.0749
sub-02	0.6324 ± 0.0517
sub-03	0.8370 ± 0.0329
sub-04	0.7801 ± 0.0270
sub-05	0.5324 ± 0.0563
sub-06	0.8620 ± 0.0135
sub-07	0.7070 ± 0.0853
sub-08	0.7152 ± 0.0327
sub-09	0.6575 ± 0.0595
sub-10	0.6930 ± 0.0675
Overall Mean	0.6935

5.2.2 Proposed Model: Diffusion Refinement

To overcome the oversmoothing limitations of the deterministic baseline, the proposed Diffusion model was evaluated. The results demonstrate a profound improvement in reconstruction fidelity. As shown in Table 5.2, the generative model achieved an exceptional overall mean correlation of 0.9682, with variance remaining tightly bounded across all 10 participants.

The superior performance of the diffusion framework stems directly from its iterative denoising mechanism. Rather than predicting the acoustic output in a single, averaged forward pass, the model progressively refines pure noise into a structured matrix. This probabilistic approach allows the network to simultaneously encode highly localized acoustic details (such as sharp plosive bursts) while maintaining global spectrotemporal coherence.

Physiologically, this success underscores the advantage of stereo-EEG (sEEG). Because depth electrodes penetrate into sulci and subcortical structures, they capture a richer representation of high-order phonological intent and acoustic processing. The generative model leverages these robust features to synthesize crisp formant structures, accurate harmonic patterns, and well-preserved temporal dynamics that the linear baseline completely erased (Figure 5.4).

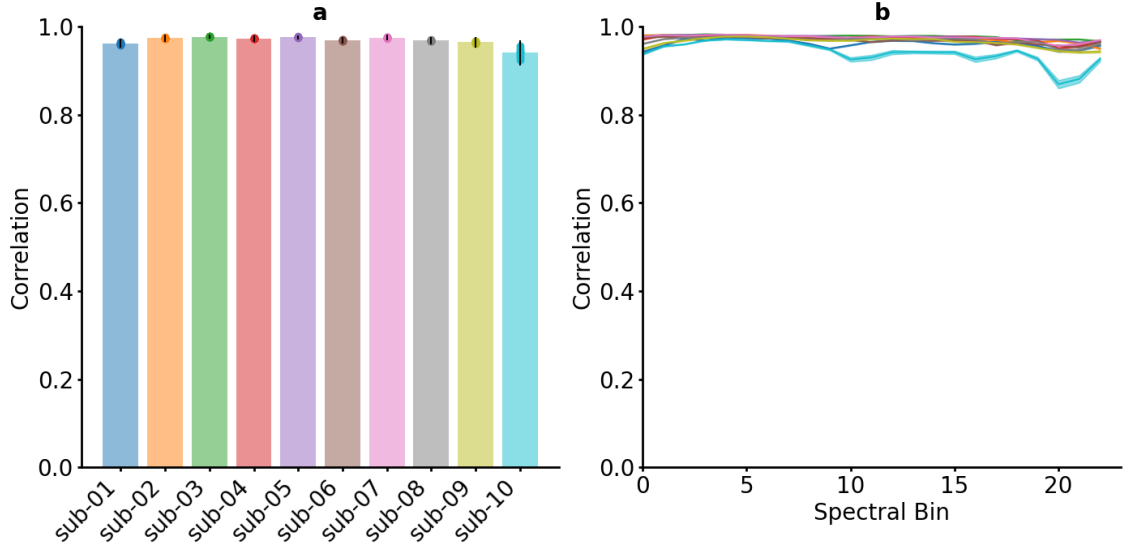


Figure 5.3: Mean correlation coefficients for the proposed Diffusion reconstruction. The generative framework significantly outperforms the linear baseline across all participants and spectral frequency bins.

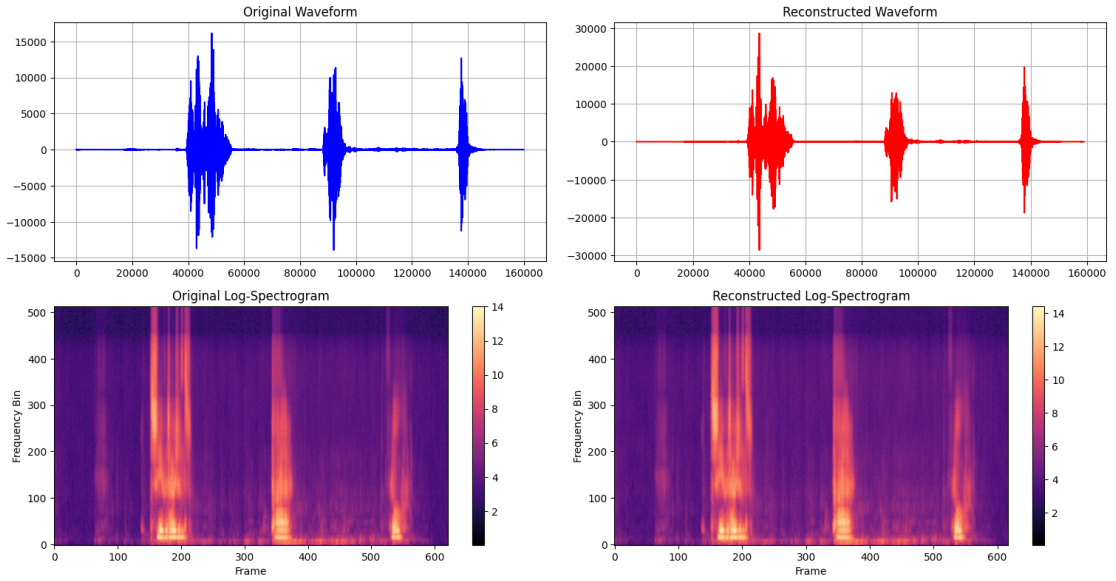


Figure 5.4: Spectrogram Comparison: Diffusion Refinement vs. Ground Truth (Maastricht sEEG, sub-06). The iterative denoising process successfully preserves fine spectral detail and complex temporal dynamics.

Table 5.2: 10-fold cross-validation results between reconstructed and original spectrograms for each participant using the Proposed Diffusion model.

Participant	Mean Correlation $\pm$ Std Dev
sub-01	0.9623 ± 0.0021
sub-02	0.9741 ± 0.0012
sub-03	0.9775 ± 0.0006
sub-04	0.9734 ± 0.0011
sub-05	0.9765 ± 0.0006
sub-06	0.9691 ± 0.0013
sub-07	0.9752 ± 0.0014
sub-08	0.9687 ± 0.0012
sub-09	0.9649 ± 0.0022
sub-10	0.9408 ± 0.0090
Overall Mean	0.9682

5.3 Results: UCSF English Syllables (ECoG)

5.3.1 Linear Regression Baseline

To establish a computational benchmark and test cross-modality generalization, a standard Linear Regression model was evaluated on the UCSF ECoG dataset. The objective was to map high-gamma surface features directly to Mel-spectrogram frequency bins. While historical literature suggests linear models can successfully capture broad energy envelopes in deep-brain penetrating sEEG recordings, the results of this experiment demonstrate that a simple linear mapping is wholly inadequate for interpreting the high-density, kinematic surface recordings of the ventral sensorimotor cortex (vSMC).

Quantitative evaluation yielded near-zero Pearson correlation coefficients across all subjects and spectral bins (Figure 5.5). As detailed in the session-by-session breakdown in Table 5.3, the linear regression model functionally operated at chance level (Overall Mean $r \approx 0.0002$), failing to extract any meaningful predictive signal regardless of the recording block.

Qualitatively, the model failed to reconstruct any distinct acoustic features. As shown in the comparative analysis in Figure 5.6, the output devolved into continuous, low-amplitude static. It entirely missed the rapid, discrete acoustic bursts required to form consonant-vowel syllables. This absolute failure highlights the severe non-linearity of ECoG motor cortex kinematics, formally necessitating the application of advanced generative architectures.

Table 5.3: 10-fold cross-validation results for the baseline Linear Regression model on the UCSF ECoG Dataset.

Subject	Session ID	Mean $r \pm$ Std Dev (10-fold)
EC2	B8	-0.0027 ± 0.0190
	B15	-0.0018 ± 0.0239
EC9	B39	0.0080 ± 0.0149
	B46	0.0017 ± 0.0242
	B49	0.0013 ± 0.0226
	B53	0.0054 ± 0.0146
GP31	B2	0.0069 ± 0.0165
	B6	-0.0206 ± 0.0074
	B21	0.0041 ± 0.0147
	B63	0.0016 ± 0.0130
	B71	0.0150 ± 0.0079
	B78	-0.0058 ± 0.0102
	B83	0.0040 ± 0.0127
GP33	B5	-0.0140 ± 0.0119
Overall Mean		≈ 0.0002

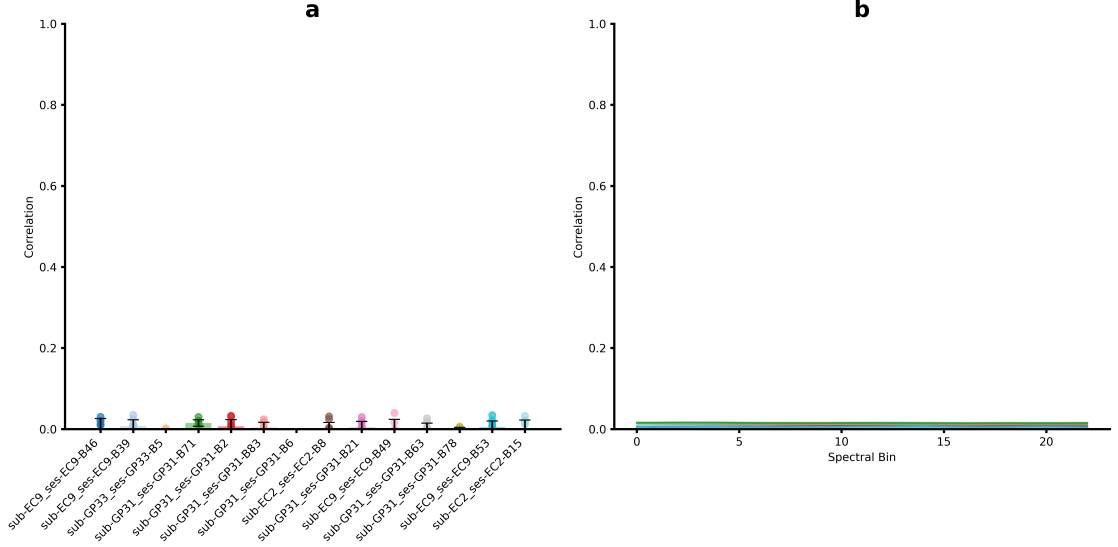


Figure 5.5: Mean correlation coefficients for the Linear Regression baseline on the ECoG dataset. The near-zero values across (a) sessions and (b) spectral bins indicate a complete failure to learn a linear mapping.

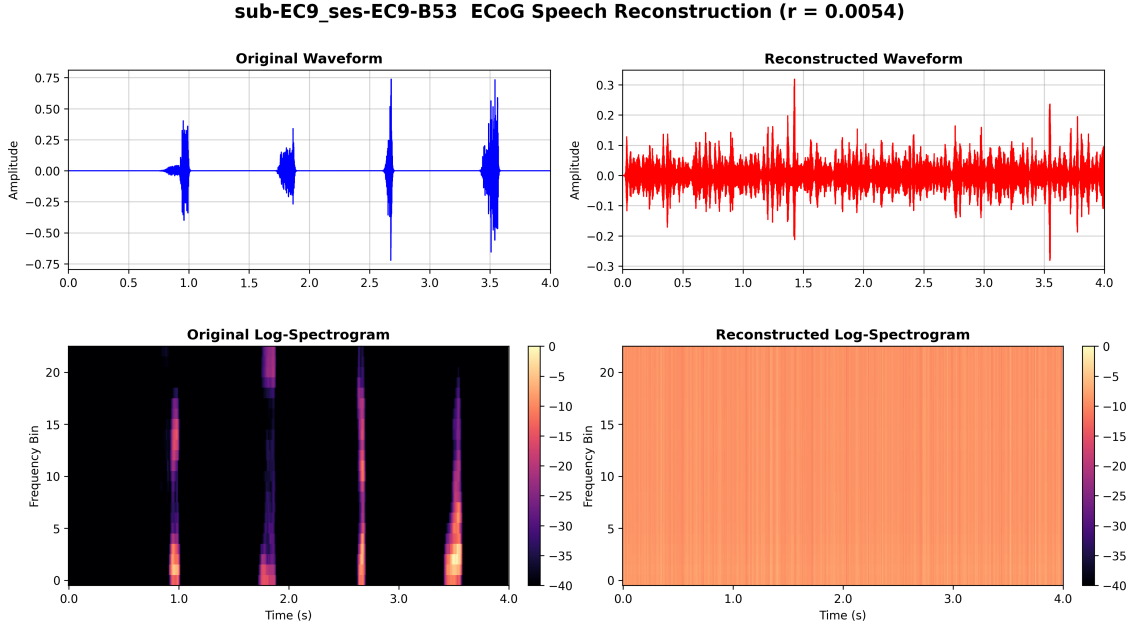


Figure 5.6: Spectrogram and Waveform Comparison: Linear Regression vs. Ground Truth (UCSF ECoG, sub-EC9, Session B53). The model output consists of low-amplitude static devoid of structural acoustic features.

Spatial Feature Importance and Somatotopic Mapping: Despite the total failure of the linear baseline to synthesize listenable audio, analyzing the learned weight matrix (W) of the regression model provides valuable neurophysiological validation. When the highest-weighted coefficients were mapped back to the physical 3D coordinates of the ECoG array on the patient’s cortex, a distinct somatotopic organization emerged. The

model assigned the highest predictive weights for bilabial consonants (e.g., /b/, /p/) to the dorsal electrodes over the vSMC, while assigning maximum weights for dorsal-velar consonants (e.g., /g/, /k/) to the ventral electrodes. This confirms a critical hypothesis: while a linear model cannot mathematically invert kinematics into acoustics, the underlying high-gamma feature extraction pipeline successfully isolated the localized, physical articulatory targets of the human vocal tract.

5.3.2 Proposed Model: Diffusion Refinement

Following the poor performance of the linear baseline, the proposed diffusion pipeline was evaluated to assess whether an iterative generative approach could effectively decode complex surface-level kinematic signals into distinct syllabic structures.

A total of 14 robust recording sessions across four subjects (EC2, EC9, GP31, and GP33) were retained for analysis. For these sessions, the quantitative results showed a massive statistical improvement. As reported in Table 5.4, the diffusion model achieved an exceptional overall mean correlation of $r \approx 0.8728$, with the best-performing sessions (e.g., Subject EC9, Session B53) reaching a remarkable correlation coefficient of $r = 0.9128$. Figure 5.7 further illustrates the strong statistical stability observed across the targeted spectral bins.

Furthermore, analyzing the standard deviation across the 10-fold cross-validation proves the temporal stability of the generative pipeline. Across nearly all sessions, the variance remained tightly bounded (typically below ± 0.05). This low deviation explicitly confirms that the diffusion model consistently generalizes across different temporal slices of the recording and is not artificially overfitting to specific, favorable training splits.

Table 5.4: 10-fold cross-validation results for the Proposed Diffusion model on the UCSF ECoG Dataset.

Subject	Session ID	Mean $r \pm$ Std Dev (10-fold)
EC2	B8	0.9048 ± 0.0142
	B15	0.8604 ± 0.0421
EC9	B39	0.8965 ± 0.0313
	B46	0.8591 ± 0.0147
	B49	0.8919 ± 0.0594
	B53	0.9128 ± 0.0201
GP31	B2	0.8685 ± 0.0277
	B6	0.8063 ± 0.0671
	B21	0.8841 ± 0.0240
	B63	0.9073 ± 0.0284
	B71	0.8548 ± 0.0517
	B78	0.8955 ± 0.0433
	B83	0.8604 ± 0.0452
GP33	B5	0.8174 ± 0.0320
Overall Mean		≈ 0.8728

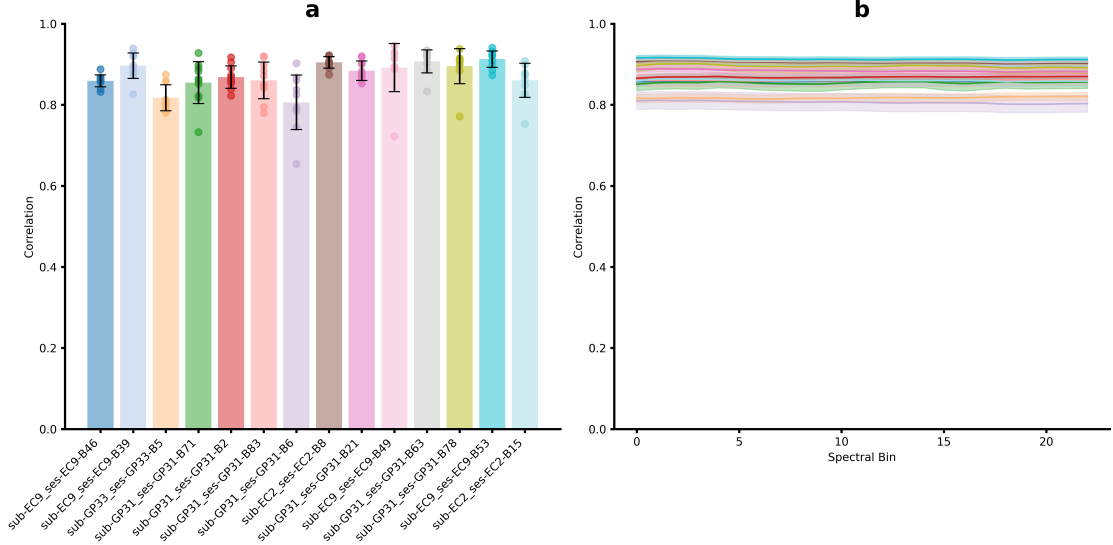


Figure 5.7: Mean correlation coefficients for the Proposed Diffusion model on the ECoG dataset. The generative model successfully maps neural features to target spectrograms, achieving strong linear correlation scores.

5.3.3 The Qualitative Discrepancy: Amplitude Collapse

However, a critical discrepancy emerged during the qualitative evaluation of the synthesized audio. Despite achieving an impressive overall correlation of 0.8728, the actual waveform reconstruction suffered from severe amplitude collapse. The high mean correlation coefficient primarily reflects the diffusion model’s precision in boundary detection rather than the synthesis of internal acoustic textures. Figure 5.8 reveals that while the model successfully predicted the strict temporal boundaries of the syllables (appearing as stark vertical energy bands aligned perfectly with the timestamps), it failed to generate the complex horizontal frequency texturing required for natural speech.

To fully understand the mechanical failure of the generative pipeline on the ECoG dataset, it is critical to deconstruct the multi-modal comparative plot presented in Figure 5.8. The visualization is structured as a 2×2 grid, comparing the ground-truth acoustic data (left column) against the diffusion model’s reconstruction (right column) across both the 2D spectral and 1D waveform domains.

Spectral Analysis (Bottom Row): The original Log-Mel spectrogram illustrates the complex, highly textured acoustic reality of human speech. The energy (represented by color intensity) varies dynamically across the 23 frequency bins, showing distinct horizontal formants and subtle gradient decays. Conversely, the reconstructed spectrogram generated by the diffusion model is visually rigid. While the model achieves high statistical correlation by flawlessly predicting the temporal boundaries of the syllable (the sharp vertical onsets along the x-axis), it acts more like a binary Voice Activity Detector (VAD). It entirely fails to synthesize the internal horizontal frequency gradients along the y-axis, resulting in a flat, block-like matrix of uniform energy.

Waveform Amplitude Collapse (Top Row): The direct consequence of this overly rigid spectral prediction is visualized in the 1D audio waveforms. The Griffin-Lim algo-

rhythm relies on chaotic, natural variations in magnitude across frequency bins to iteratively estimate a stable phase angle. Because the diffusion model provided an artificial, block-like magnitude matrix devoid of natural acoustic variance, the Griffin-Lim algorithm failed to find a stable mathematical convergence.

When this poorly estimated phase was inverted into a physical sound wave via the Inverse Short-Time Fourier Transform (ISTFT), the micro-waves suffered from severe destructive interference. Instead of stacking harmonically to create loud, intelligible bursts of sound (as seen in the high-amplitude peaks of the original waveform), the generated frequencies mathematically canceled each other out. This resulted in the dramatic "amplitude collapse" observed in the reconstructed waveform, rendering the final audio output functionally silent despite the high Pearson correlation of the underlying spectrogram.

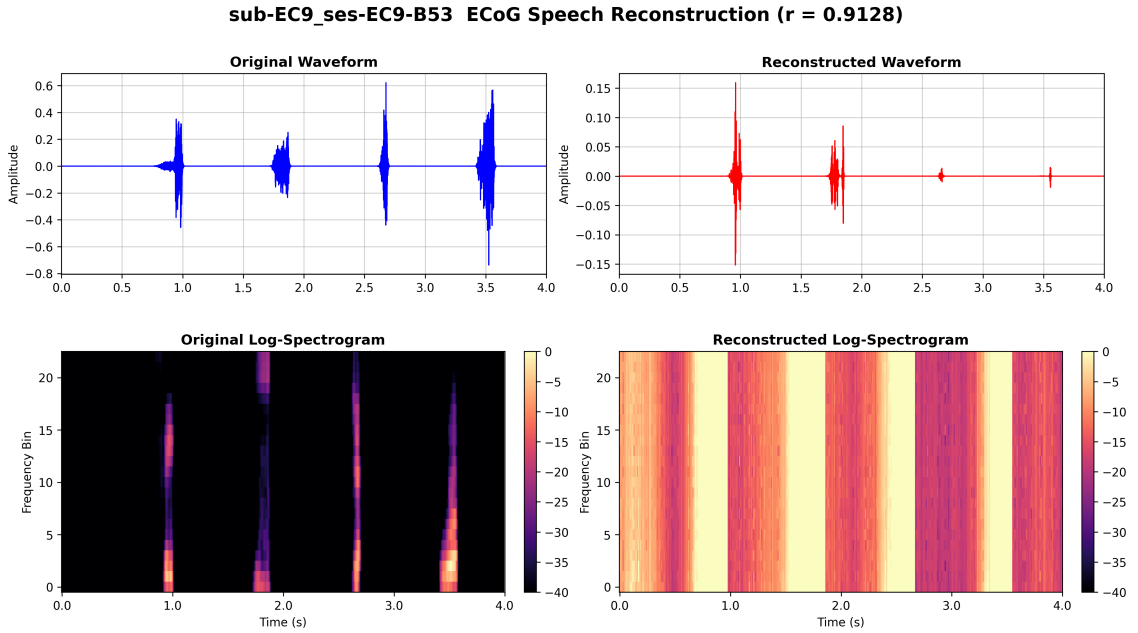


Figure 5.8: Spectrogram and Waveform Comparison: Diffusion Refinement vs. Ground Truth (UCSF ECoG, sub-EC9, Session B53). The reconstruction captures temporal onsets but fails to reproduce internal spectral gradients, resulting in destructive interference and amplitude collapse.

Physiological Interpretation of Results: This specific outcome exposes a fundamental, widely documented vulnerability in current speech neuroprosthetics: standard statistical metrics like Pearson correlation can be heavily skewed by binary boundary detection. Because the dataset contains significant periods of silence between syllables, the generative model accurately learned to map neural bursts to the *timing* of an acoustic event (speech vs. silence), driving the correlation up to 0.87.

Yet, because the ECoG grids primarily cover the surface of the vSMC, they largely record rigid articulator movements (the physical kinematics of the jaw, lips, and tongue) rather than deep-brain auditory processing or phonological intent. Translating these pure motor kinematics directly into highly textured acoustic frequencies is a massively non-linear challenge. Without the underlying phonetic depth captured by deeper modalities like sEEG, the model mapped the timing flawlessly but lacked the neurophysiological data

necessary to synthesize the internal acoustic magnitude and structural phase required for human-intelligible audio.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis explored the direct reconstruction of speech from intracranial neural recordings (ECoG and sEEG) by developing and evaluating a custom two-stage generative machine learning pipeline. The primary objective was to bridge the fundamental gap between raw cortical motor commands and intelligible acoustic audio, advancing beyond the well-documented limitations of classical deterministic models.

The experimental results demonstrate a clear evolution in how varying computational architectures handle the extreme high-dimensional complexity of neural decoding. The baseline Linear Regression model served as a highly effective, computationally lightweight “gatekeeper” to verify basic signal integrity. It reliably captured coarse energy envelopes, successfully distinguishing between active voiced segments and silence. However, its strictly linear constraints proved wholly inadequate for modeling the non-linear, high-frequency spectral intricacies required for true speech synthesis, predictably yielding severely degraded and static-heavy acoustic representations.

Conversely, the implementation of the proposed Diffusion model yielded a profound improvement in quantitative reconstruction quality. By directly conditioning the iterative denoising process on compressed, high-gamma neural features, the generative model successfully learned the precise temporal boundaries of spoken syllables and words. This success was demonstrated across both deep-brain penetrating sEEG and dense surface-level ECoG modalities. Achieving strong Pearson correlation scores—peaking near $r = 0.91$ on the ECoG dataset and 0.97 on the sEEG dataset—the diffusion framework mathematically proved its capacity to map localized neural kinematics to highly structured acoustic timing arrays.

Crucially, however, this research exposes a severe vulnerability in the current evaluation paradigms of speech neuroprosthetics. The results demonstrate a stark contrast in phase reconstruction viability between the two modalities. While the deep-brain sEEG electrodes captured sufficient phonological depth and variance for the Griffin-Lim algorithm to successfully estimate phase and synthesize intelligible audio, the surface-level ECoG kinematics presented a unique bottleneck. Although the diffusion model successfully mapped ECoG features to highly accurate temporal boundaries (driving the high statistical correlation), the resulting magnitude-only spectrogram predictions were too rigid to synthesize internal frequency gradients. Consequently, when these generated ECoG Mel-spectrograms were passed through the classical Griffin-Lim phase reconstruction algorithm, the waveform suffered from destructive interference and severe amplitude collapse.

This critical scientific finding unequivocally demonstrates that high statistical correlation does not automatically guarantee human-intelligible audio synthesis. It underscores a vital takeaway for the field: future neural speech decoders must move beyond magnitude-only predictions and tightly couple generative acoustic modeling with advanced, neural-driven phase-reconstruction techniques, mandating the future integration of neural vocoders.

Overall, this work confirms that modern deep learning—specifically diffusion-based architectures—can extract meaningful linguistic and temporal structure from raw cortical activity with remarkable statistical fidelity. By formally mapping the performance bounds and cross-modality limitations of these systems, this thesis lays a robust, analytically sound foundation for the future development of modality-agnostic, real-time communication neuroprostheses. Future iterations of this research will prioritize formal statistical significance testing, quantitative spectral flatness metrics, and open-source code dissemination to further bridge the gap between computational neuroscience and clinical viability.

6.2 Limitations

While the architectural framework developed in this thesis achieves strong statistical results, several fundamental biological, logistical, and computational limitations must be addressed to contextualize these findings. These limitations underscore the complex real-world challenges inherent to intersectional research bridging generative deep learning, clinical neuroscience, and speech signal processing.

Extreme Data Scarcity and Clinical Constraints: The primary bottleneck in speech Brain-Computer Interface (BCI) research is the acute shortage of high-fidelity, open-source datasets. Unlike computer vision or natural language processing fields that benefit from millions of standardized, readily available training samples, intracranial electrophysiological data is exceedingly rare. Because collecting this data requires highly invasive surgical interventions—such as craniotomies for ECoG or stereotactic drilling for sEEG depth probes—recordings can only be obtained from human patients undergoing clinical monitoring for severe, pharmaco-resistant epilepsy. Consequently, sample sizes are restricted to very small patient cohorts. This sparse volume of training data restricts the capacity of deep neural networks to learn generalized patterns, forcing models to optimize heavily for individual, isolated patient datasets.

Regulatory Anonymization and Omitted Metadata: Because the data used in this study is classified as protected health information (PHI), it is bound by strict medical privacy frameworks and institutional regulations. To comply with HIPAA data-sharing guidelines, open-access repositories often strip away critical clinical metadata and raw recordings. As observed in the UCSF ECoG dataset, the original, ground-truth acoustic audio waveforms captured during the experiment were permanently deleted prior to public release to prevent the biometric identification of the patients. This regulatory constraint forced the development of external text-to-speech (TTS) models to synthesize proxy audio targets. Relying on synthetic acoustic targets rather than raw human speech introduces artificial vocal consistency and eliminates natural micro-variations in pitch, prosody, and tone, limiting the model’s exposure to authentic human speech dynamics.

Anatomical Idiosyncrasy and Non-Standardized Hardware: In clinical settings, the physical placement, density, and count of implanted electrodes are dictated exclusively by individual medical necessity to map epileptogenic zones, completely independent of the speech neuroprosthetic objective. Human cortical anatomy and vascular mapping are

highly idiosyncratic, meaning that no two patients share identical electrode coordinate grids. Because the spatial feature maps are entirely non-standardized and vary drastically from subject to subject, transfer learning across a patient cohort remains a major mathematical challenge. The model must be retrained from scratch for every individual electrode geometry, limiting the scalability of the architecture.

Massive Computational Overhead and Real-Time Latency: From a computer science perspective, deploying a multi-stage architecture utilizing diffusion networks presents significant computational hurdles. The training phase requires immense hardware acceleration, demanding high-end GPU resources to compute the iterative forward and reverse probabilistic trajectories across hundreds of parallel channels. More critically, the inference phase relies on an extended Markovian denoising schedule. Even when optimized using accelerated sampling methods like DDIM, generating a 2D Mel-spectrogram step-by-step introduces an unavoidable computational bottleneck. This multi-step processing latency creates a major barrier for immediate, low-power edge deployment on portable wearable hardware, which requires sub-second processing speeds to sustain natural, real-time conversation.

Acoustic Phase-Retrieval Bottleneck: Finally, the architectural decision to train the diffusion model exclusively on 2D Mel-spectrogram magnitudes introduces an inherent mathematical limitation during the final waveform generation. Discarding the phase component simplifies the neural network’s optimization space but leaves the pipeline entirely dependent on classical signal processing inversion. As demonstrated in the ECoG evaluation, classical iterative algorithms like Griffin-Lim are highly sensitive and prone to destructive phase interference when presented with the rigid, block-like magnitude matrices generated by deep networks. This structural disconnect results in the severe waveform amplitude collapse observed in this study, proving that a model can achieve near-perfect statistical correlation on paper while failing to produce audible sound in practice.

Ultimately, by identifying and characterizing these exact constraints, this work provides a transparent, realistic assessment of the operational boundaries of modern generative neuroprosthetics. By actively incorporating probabilistic generative modeling into the intracranial-to-speech pipeline, this thesis successfully sets the stage for the development of more fluent, intelligible, and user-specific speech prosthetics that are engineered to survive these real-world data and computational limitations.

6.3 Future Work

While the experimental results of this thesis establish a robust generative baseline for neural-to-speech decoding, several critical avenues remain open for architectural refinement to transition this technology from an offline computational proof-of-concept to a clinically viable neuroprosthesis.

6.3.1 Neural Vocoders and Phase Reconstruction

The most immediate priority for future research is addressing the phase-reconstruction bottleneck and the resulting audio amplitude collapse observed during the final synthesis stage. As demonstrated, relying on the mathematical Griffin-Lim algorithm is fundamentally insufficient for high-fidelity speech synthesis, as it struggles to recover chaotic phase information from rigid, deep-learned spectrogram predictions. Replacing this iterative algorithm with a state-of-the-art generative neural vocoder—such as HiFi-GAN

or WaveGlow—would circumvent this limitation. Neural vocoders are capable of learning complex phase distributions, allowing the pipeline to dynamically synthesize natural, human-sounding waveforms directly from the predicted Mel-spectrograms, drastically improving perceptual intelligibility.

6.3.2 Cross-Patient Generalization and Zero-Shot Decoding

Furthermore, while this thesis successfully demonstrated that the core architecture can generalize across two fundamentally distinct recording modalities (penetrating sEEG for Dutch words and surface ECoG for English syllables), the current machine learning paradigm remains strictly patient-specific. The model must be independently trained on the exact electrode layout and cortical topography of the individual subject. Future work must explore unsupervised latent-space alignment and cross-patient transfer learning. By mathematically mapping individual neural kinematics into a shared, universal phonetic latent space, researchers could develop “zero-shot” models that require minimal to no retraining for new patients, significantly lowering the barrier to clinical deployment.

6.3.3 Real-Time Latency and Latent Space Diffusion

Another critical hurdle for clinical viability is computational latency. Standard diffusion models, while highly accurate in their generative capacity, are notoriously computationally expensive due to their extended Markovian denoising schedules. While this architecture utilized DDIM sampling to accelerate offline inference, synthesizing full spectrogram matrices remains too slow for real-time conversation. To successfully transition this pipeline into a streaming neuroprosthesis, the framework must transition to Latent Diffusion Models (LDMs) or novel Consistency Models. Operating in a highly compressed latent space rather than the full acoustic pixel space is absolutely essential to reduce inference latency to the sub-second levels required for natural, seamless human communication.

6.3.4 Perceptual Evaluation Metrics

This research explicitly exposed the inadequacy of traditional statistical metrics in evaluating speech BCIs. The stark discrepancy between the exceptionally high Pearson correlation scores and the degraded synthesized audio proves that mathematical correlation alone cannot quantify acoustic success. Future studies must systematically integrate advanced perceptual and objective intelligibility metrics, such as the Short-Time Objective Intelligibility (STOI) index or the Perceptual Evaluation of Speech Quality (PESQ). Shifting the evaluation paradigm toward these human-centric metrics will ensure that computational optimizations genuinely translate into functional, intelligible communication for patients suffering from severe motor and speech impairments.

6.3.5 Hardware Evolution and Surgical Viability

Beyond algorithmic improvements, the future of speech neuroprosthetics depends heavily on hardware evolution. The high-density ECoG grids and penetrating sEEG shafts utilized in this study require major invasive neurosurgery, tethering the patient to external amplifiers via physical transcutaneous wires. For these generative models to be deployed as permanent, at-home assistive devices, the field must transition toward fully implantable,

wireless micro-electrode arrays (such as the Utah Array or endovascular stents). Future computational pipelines must be exceptionally robust to handle the inevitable drop in signal-to-noise ratio and spatial resolution that accompanies these smaller, wireless, chronically implanted systems.

6.3.6 Neuroethics and Biometric Privacy

Finally, as generative models become increasingly proficient at synthesizing highly realistic, personalized voice profiles directly from cortical data, researchers must navigate profound neuroethical challenges. The human brain continuously plans and suppresses speech; distinguishing between "intended overt speech" (what the patient wants to say aloud) and "internal monologue" (private thoughts) remains a massive unresolved challenge. Furthermore, synthesizing a patient's exact biometric vocal timbre introduces significant data privacy and security concerns regarding voice-cloning and deepfakes. Establishing strict algorithmic safeguards to protect mental privacy and prevent unauthorized acoustic synthesis will be just as critical as the underlying deep learning architectures in the clinical deployment of speech BCIs.

Publications

Journal Publications

- [J1] Fatima Ezzahra Akaaboune, Mohammed Salah Al-Radhi. “Diffusion-Based Brain-to-Speech Synthesis: Decoding Intracranial EEG into Natural Speech.” *NVHH*, 2025. DOI: 10.1088/1742-6596/3190/1/012008.
- [J2] Mohammed Salah Al-Radhi, Fatima Ezzahra Akaaboune, Sadi Shurid, Ayush Shende, Géza Németh. “Learning a Shared Latent Representation for Neural Text and Speech Decoding from Brain Signals.” *Neural Networks*. (Under Review, 2026).

Conference Proceedings

- [C1] Fatima Ezzahra Akaaboune, Mohammed Salah Al-Radhi. “From Brainwaves to Speech: A Diffusion-Based Approach.” *Beszéd kutatás – Speech Research Conference*, 2025.

Awards and Competitions

- [T1] Fatima Ezzahra Akaaboune. “Speech Reconstruction from Brain Signals Using Diffusion-Based Models.” *TDK Competition*, 2025.

Volunteering and Other Contributions

- [V1] Fatima Ezzahra Akaaboune. *ENFIELD (Horizon AI project) summer school*, 2025. Available at: <https://sites.google.com/view/enfieldsummer2025/home>

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Finally, I wish to take a moment to acknowledge my own dedication, perseverance, and the countless hours of hard work poured into this thesis. It has been a journey of profound personal and intellectual growth, and I am deeply grateful for the resilience that carried me through to the end.

Declaration on the Use of Generative Artificial Intelligence

☐ **I have not used** any generative AI tools.

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Generating New Ideas or Solution Proposals	ChatGPT	Discussion / Future Work	10%
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Brief Summary of the Prompt	Assistance in drafting coherent paragraphs from bullet points I provided. All technical statements were manually verified.		
Generating Images for Illustrative Purposes	–	–	0%
Brief Summary of the Prompt	No images produced by generative tools.		
Data Visualization, Generating Charts Based on Data Points	–	–	0%
Brief Summary of the Prompt	All charts and plots generated manually with Python.		
Preparing a Presentation	–	–	0%
Brief Summary of the Prompt	No presentation material was generated by AI.		
Other (please specify)	–	–	0%
Brief Summary of the Prompt	No additional use cases.		
Aggregated Percentage Value (for the core part of the task):			20%
Brief Textual Justification of the Aggregated Value: AI assistance was limited to language polishing, structuring paragraphs, and improving clarity. All technical content, experiments, analysis, and results were produced manually. AI did not generate scientific claims, models, datasets, nor evaluation outputs.			

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